

Aptitude Mastery Guide

Every topic and sub-topic explained: what it is, how to identify it, how to solve it, with worked examples

Quantitative Aptitude · Logical Reasoning · Verbal Ability · Data Interpretation ·
Technical Aptitude

How to use this guide

- Every sub-topic follows the same four-part pattern: **What it is** (a one-line definition), **How to identify it** (the wording or pattern that tells you which method to reach for), **How to solve it** (the method, with the key formulas), and **two worked examples**.
- Learn to **identify first**. Most lost marks in aptitude tests come from choosing the wrong method, not from calculation slips. Read the identify bullets before the solve steps.
- Figure-based reasoning questions (mirror images, paper folding, figure series) are described in words and symbols here. The method is the same on the real figures; practise those on paper too.
- Suggested routine: read one sub-topic, cover the solutions, solve both examples yourself, then compare. Then do 10 fresh questions from any practice book on the same sub-topic.
- Time targets for most company placement tests: quantitative about 1 minute per question, reasoning about 1 minute, verbal 45 to 60 seconds, data interpretation 1.5 to 2 minutes. Skip anything that crosses double the target and return to it at the end.
- Formulas are written in plain text: x^2 means x squared, a_n means the n-th term, $\times$ is multiplication, $\div$ is division and $\sqrt{\quad}$ is the square root.

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Part 1: Quantitative Aptitude

Numerical ability: the largest section in most placement tests. Speed comes from recognising the problem type in the first few seconds.

1.1 Number System

1.1.1 Types of numbers

What it is: Numbers are grouped by their properties: natural (1, 2, 3, ...), whole (0, 1, 2, ...), integers (... , -2, -1, 0, 1, 2, ...), rational (can be written as p/q), irrational (non-terminating, non-repeating decimals such as $\sqrt{2}$ and π), prime (exactly two factors) and composite (more than two factors).

How to identify it

- The question uses words like prime, composite, co-prime, rational, irrational, perfect number, consecutive.
- "How many primes between...", "smallest composite number", "sum of first n natural numbers".

How to solve it

- To test if n is prime, divide by every prime up to $\sqrt{n}$. If none divides it, n is prime.
- 1 is neither prime nor composite. 2 is the only even prime.
- Sum of first n natural numbers = $n(n+1)/2$. Sum of first n odd numbers = n^2 . Sum of first n even numbers = $n(n+1)$.
- Co-prime numbers have HCF 1 and need not be prime themselves (8 and 15 are co-prime).

Example 1. Is 221 prime?

Solution. $\sqrt{221}$ is about 14.9, so test primes up to 13.

$221 \div 13 = 17$ exactly, so $221 = 13 \times 17$. It is composite.

Example 2. Find the sum of all odd numbers from 1 to 49.

Solution. Odd numbers from 1 to 49 are the first 25 odd numbers, because $49 = 2 \times 25 - 1$.

Sum = $25^2 = 625$.

1.1.2 Divisibility rules

What it is: Quick tests that tell whether a number divides another exactly, without carrying out the division.

How to identify it

- "Find the digit * so that the number is divisible by...", "which of these is divisible by..."
- "Smallest number to add or subtract to make it divisible by..."

How to solve it

- 2: last digit even. 3: digit sum divisible by 3. 4: last two digits divisible by 4. 5: last digit 0 or 5.
- 8: last three digits divisible by 8. 9: digit sum divisible by 9. 10: last digit 0.
- 11: (sum of digits in odd places) – (sum of digits in even places) is 0 or a multiple of 11.
- 7: remove the last digit, subtract twice that digit from the rest, repeat until small.
- For composite divisors test co-prime parts: $6 = 2 \times 3$, $12 = 3 \times 4$, $72 = 8 \times 9$.

Example 1. Find the digit x so that $4x56$ is divisible by 9.

Solution. Digit sum = $4 + x + 5 + 6 = 15 + x$.

The next multiple of 9 after 15 is 18, so $x = 3$. Check: $4356 \div 9 = 484$.

Example 2. Is 918082 divisible by 11?

Solution. Digits from the left: 9, 1, 8, 0, 8, 2.

Odd places (1st, 3rd, 5th): $9 + 8 + 8 = 25$. Even places: $1 + 0 + 2 = 3$.

Difference = 22, a multiple of 11, so yes. ($918082 \div 11 = 83462$.)

1.1.3 HCF & LCM

What it is: The HCF is the greatest number that divides every given number exactly. The LCM is the smallest number that every given number divides exactly.

How to identify it

- HCF situations (splitting): greatest number that divides, largest tile, longest rope that measures exactly, maximum number of equal groups.
- LCM situations (meeting again): bells ringing together, runners meeting at the start, smallest number divisible by all, "after how much time together again".

How to solve it

- Prime factorise. HCF = product of the lowest powers of common primes. LCM = product of the highest powers of all primes.
- For two numbers: $\text{HCF} \times \text{LCM} = \text{product of the numbers}$.
- Greatest number that divides a, b, c leaving remainders $r_1, r_2, r_3 = \text{HCF of } (a - r_1), (b - r_2), (c - r_3)$.
- Smallest number leaving the same remainder r on division by several numbers = $\text{LCM} + r$.
- Fractions: $\text{HCF} = \text{HCF of numerators} \div \text{LCM of denominators}$. $\text{LCM} = \text{LCM of numerators} \div \text{HCF of denominators}$.

Example 1. Find the greatest number that divides 285 and 1249 leaving remainders 9 and 7 respectively.

Solution. Subtract the remainders: $285 - 9 = 276$ and $1249 - 7 = 1242$.

$\text{HCF}(276, 1242)$: $1242 = 4 \times 276 + 138$, and $276 = 2 \times 138$, so the HCF is 138.

138 is larger than both remainders, so it is valid.

Example 2. Three bells ring every 12, 15 and 20 minutes. They ring together at 8:00 a.m. When next together?

Solution. $12 = 2^2 \times 3$, $15 = 3 \times 5$, $20 = 2^2 \times 5$.

$\text{LCM} = 2^2 \times 3 \times 5 = 60$ minutes, so they ring together next at 9:00 a.m.

1.1.4 Factors & multiples

What it is: Factors of a number divide it exactly; multiples are obtained by multiplying it by whole numbers. Counting factors is done from the prime factorisation.

How to identify it

- "How many factors/divisors", "sum of all factors", "product of factors".
- "In how many ways can N be written as a product of two numbers", "number of odd factors", "perfect square".

How to solve it

- Write $N = a^p \times b^q \times c^r$ (primes a, b, c).
- Number of factors = $(p+1)(q+1)(r+1)$.
- Sum of factors = $(a^{p+1} - 1)/(a - 1) \times (b^{q+1} - 1)/(b - 1) \times (c^{r+1} - 1)/(c - 1)$.
- Ways to write N as a product of two different factors = $(\text{number of factors}) \div 2$ if N is not a perfect square, or $(\text{number of factors} - 1) \div 2$ if it is.
- N is a perfect square exactly when every exponent is even (so it has an odd number of factors).
- For odd factors, ignore the prime 2 in the count.

Example 1. How many factors does 360 have, and what is their sum?

Solution. $360 = 2^3 \times 3^2 \times 5$.

Number of factors = $4 \times 3 \times 2 = 24$.

Sum = $(2^4 - 1)/1 \times (3^3 - 1)/2 \times (5^2 - 1)/4 = 15 \times 13 \times 6 = 1170$.

Example 2. In how many ways can 144 be written as a product of two different factors?

Solution. $144 = 2^4 \times 3^2$, so it has $5 \times 3 = 15$ factors. It is a perfect square.

Ways = $(15 - 1) \div 2 = 7$. They are $1 \times 144, 2 \times 72, 3 \times 48, 4 \times 36, 6 \times 24, 8 \times 18, 9 \times 16$.

1.1.5 Remainders

What it is: Finding what is left when a number, often a huge power or product, is divided by another number.

How to identify it

- "Remainder when ... is divided by ...", especially with large powers, products, or factorial sums.
- "Last two digits" is the remainder on division by 100.

How to solve it

- Remainder of a sum or product = sum or product of the individual remainders, reduced again.
- Use negative remainders to shrink numbers: $99 \equiv -1 \pmod{100}$, $10 \equiv -1 \pmod{11}$.
- For powers, list the remainders until they repeat, then use the exponent modulo the cycle length.
- Fermat: if p is prime and a is not a multiple of p , then a^{p-1} leaves remainder 1 on division by p .
- Wilson: $(p-1)!$ leaves remainder $p-1$ on division by p , for prime p .
- $n!$ is divisible by every number up to n , so most terms of a factorial sum vanish.

Example 1. Find the remainder when 2^{100} is divided by 7.

Solution. Powers of 2 leave remainders 2, 4, 1 and then repeat (cycle length 3).

$100 = 3 \times 33 + 1$, so 2^{100} leaves the same remainder as 2^1 , which is 2.

Example 2. Find the remainder when $1! + 2! + 3! + \dots + 50!$ is divided by 12.

Solution. From $4! = 24$ onward, every factorial is divisible by 12.

So the remainder is that of $1! + 2! + 3! = 1 + 2 + 6 = 9$. Remainder = 9.

1.1.6 Unit digit & cyclicity

What it is: The last digit of a^n repeats in a short cycle that depends only on the last digit of a .

How to identify it

- "Unit digit of 7^{123} ", "last digit of a product or sum of powers".
- Power towers, such as 3 raised to (4 raised to 5), need care with the exponent.

How to solve it

- Digits 0, 1, 5, 6 never change. Last digit 4 and 9 cycle with length 2 (4, 6 and 9, 1).
- Last digit 2, 3, 7, 8 cycle with length 4: $2 \rightarrow 2, 4, 8, 6$; $3 \rightarrow 3, 9, 7, 1$; $7 \rightarrow 7, 9, 3, 1$; $8 \rightarrow 8, 4, 2, 6$.
- Divide the exponent by the cycle length. The remainder picks the term; remainder 0 means the last term of the cycle.
- Multiply the unit digits of the separate parts and keep only the last digit.
- Power tower: work from the top. Find the remainder of the upper exponent on division by the cycle length (4 for 2, 3, 7, 8), then use that as the exponent of the base.

Example 1. Find the unit digit of 7^{95} .

Solution. The cycle for 7 is 7, 9, 3, 1 (length 4).

$95 \div 4$ leaves remainder 3, so take the third term: 3.

Example 2. Find the unit digit of $(234)^{100} \times (567)^{101}$.

Solution. Unit digit of 4^{100} : even exponent gives 6.

Unit digit of 7^{101} : $101 \div 4$ leaves 1, so 7.

$6 \times 7 = 42$, so the unit digit is 2.

1.1.7 Factorials & trailing zeros

What it is: $n! = 1 \times 2 \times 3 \times \dots \times n$. Each trailing zero needs a factor pair 2×5 , and fives are scarcer than twos.

How to identify it

- "Number of zeros at the end of $n!$ ", "highest power of p that divides $n!$ ".
- "Largest power of 12 (or 6, 10) dividing $n!$ ".

How to solve it

- Zeros in $n! = \lfloor n/5 \rfloor + \lfloor n/25 \rfloor + \lfloor n/125 \rfloor + \dots$ (keep dividing and take whole parts).
- Highest power of a prime p in $n! = \lfloor n/p \rfloor + \lfloor n/p^2 \rfloor + \lfloor n/p^3 \rfloor + \dots$
- For a composite divisor, split into primes, find each prime's power, and take the limiting one (divide by the exponent in the divisor).

Example 1. How many zeros are at the end of 100!?

Solution. $100 \div 5 = 20$, $100 \div 25 = 4$, $100 \div 125 = 0$.

Zeros = $20 + 4 = 24$.

Example 2. Find the highest power of 6 that divides 20!.

Solution. $6 = 2 \times 3$. Power of 3 in $20! = \lfloor 20/3 \rfloor + \lfloor 20/9 \rfloor = 6 + 2 = 8$.

Power of 2 in $20! = 10 + 5 + 2 + 1 = 18$, which is more than enough.

The limiting prime is 3, so the answer is 6^8 .

1.1.8 Number base systems

What it is: A base- b system uses b digits. Common bases are 2 (binary), 8 (octal), 10 (decimal) and 16 (hexadecimal, digits 0-9 and A-F where A = 10 ... F = 15).

How to identify it

- Numbers written with a subscript base such as $(1011)_2$.
- "Convert", "in base 5", "binary addition", "hexadecimal".

How to solve it

- Base b to decimal: multiply each digit by the matching power of b and add.
- Decimal to base b : divide repeatedly by b and read the remainders from the last to the first.
- Binary to octal: group bits in threes from the right. Binary to hex: group in fours.
- Arithmetic in base b is ordinary arithmetic, but you carry when a column reaches b .

Example 1. Convert 156 to binary.

Solution. $156 \div 2 = 78$ r0, $78 \div 2 = 39$ r0, $39 \div 2 = 19$ r1, $19 \div 2 = 9$ r1, $9 \div 2 = 4$ r1, $4 \div 2 = 2$ r0, $2 \div 2 = 1$ r0, $1 \div 2 = 0$ r1.

Reading remainders from bottom to top: 10011100. Check: $128 + 16 + 8 + 4 = 156$.

Example 2. Add $(1011)_2$ and $(110)_2$ and give the answer in decimal.

Solution. $1011 + 0110$: column by column with carries gives 10001.

In decimal: $11 + 6 = 17$, and $(10001)_2 = 16 + 1 = 17$.

1.1.9 Surds & indices

What it is: Indices (powers) show repeated multiplication. Surds are irrational roots such as $\sqrt{3}$ that cannot be simplified to whole numbers or fractions.

How to identify it

- Equations with the unknown in the power: "if $4^{x+1} = 8^{x-1}$, find x ".
- Expressions with roots to simplify or rationalise.

How to solve it

- Laws: $a^m \times a^n = a^{m+n}$; $a^m \div a^n = a^{m-n}$; $(a^m)^n = a^{mn}$; $a^0 = 1$; $a^{-n} = 1/a^n$; $a^{1/n}$ is the n -th root of a .
- If the bases are equal, equate the exponents. Rewrite numbers to a common base first ($8 = 2^3$, $9 = 3^2$).
- Rationalise: multiply top and bottom by the conjugate. $1/(\sqrt{a} - \sqrt{b}) = (\sqrt{a} + \sqrt{b})/(a - b)$.
- Only like surds add or subtract: $2\sqrt{3} + 5\sqrt{3} = 7\sqrt{3}$, but $\sqrt{2} + \sqrt{3}$ stays as it is.

Example 1. Solve $4^{x+1} = 8^{x-1}$.

Solution. Write both with base 2: $2^{2(x+1)} = 2^{3(x-1)}$.

So $2x + 2 = 3x - 3$, which gives $x = 5$.

Example 2. Rationalise $1/(\sqrt{5} - \sqrt{3})$.

Solution. Multiply top and bottom by $(\sqrt{5} + \sqrt{3})$.

Result = $(\sqrt{5} + \sqrt{3})/(5 - 3) = (\sqrt{5} + \sqrt{3})/2$.

1.1.10 Logarithms

What it is: $\log_b N = x$ means $b^x = N$. A logarithm is the power to which the base must be raised to get the number.

How to identify it

- The word "log" or "log to the base", equations like $\log_2 x = 5$.
- "How many digits does 2^{50} have" (given $\log 2$).

How to solve it

- $\log(mn) = \log m + \log n$; $\log(m/n) = \log m - \log n$; $\log(m^n) = n \log m$.
- $\log_b b = 1$, $\log_b 1 = 0$, $\log_b a = 1 / \log_a b$, and change of base: $\log_b a = \log a / \log b$.
- Convert a log equation into its power form to solve it.
- Number of digits in $N = \lfloor \log_{10} N \rfloor + 1$.

Example 1. Evaluate $\log_2 64 + \log_3 81 - \log_5 125$.

Solution. $2^6 = 64$ gives 6. $3^4 = 81$ gives 4. $5^3 = 125$ gives 3.

Total = $6 + 4 - 3 = 7$.

Example 2. Given $\log_{10} 2 = 0.3010$, find the number of digits in 2^{50} .

Solution. $\log_{10}(2^{50}) = 50 \times 0.3010 = 15.05$.

Digits = $\lfloor 15.05 \rfloor + 1 = 16$.

1.2 Simplification & Approximation

1.2.1 BODMAS

What it is: The agreed order for evaluating an expression: Brackets, Orders (powers and roots), Division and Multiplication, then Addition and Subtraction.

How to identify it

- A long expression mixing brackets, powers, $\times$, $\div$, $+$ and $-$; "find the value of".

How to solve it

- Open brackets from the innermost: $()$ first, then $\{ \}$, then $[]$.
- Next powers and roots. Then division and multiplication together, working left to right.
- Finally addition and subtraction together, left to right.
- "Of" means multiplication and is done before ordinary division and multiplication.

Example 1. Evaluate $48 \div 6 \times 2 + 3^2 - (10 - 4)$.

Solution. Brackets: $10 - 4 = 6$. Orders: $3^2 = 9$.

Division and multiplication left to right: $48 \div 6 = 8$, then $8 \times 2 = 16$.

Then $16 + 9 - 6 = 19$.

Example 2. Evaluate $15 - [4 + \{6 - (3 - 1)\}] \div 2$.

Solution. Innermost: $(3 - 1) = 2$. Then $\{6 - 2\} = 4$. Then $[4 + 4] = 8$.

$8 \div 2 = 4$, and $15 - 4 = 11$.

1.2.2 Fractions & decimals

What it is: Converting between fractions and decimals, comparing them, and doing arithmetic with them, including recurring decimals.

How to identify it

- Mixed fractions, recurring decimals such as $0.(36)$, "arrange in ascending order", "which is greater".

How to solve it

- Pure recurring: $0.(ab) = ab/99$ and $0.(a) = a/9$.
- Mixed recurring: $0.a(b) = (ab - a)/90$. For example $0.1(6) = (16 - 1)/90 = 1/6$.
- Compare fractions by cross-multiplying or by converting to decimals.
- Learn: $1/3 = 0.333$, $1/6 = 0.1667$, $1/7 = 0.142857$, $1/8 = 0.125$, $1/9 = 0.111$, $1/11 = 0.0909$.

Example 1. Convert $0.(36)$ into a fraction.

Solution. $0.(36) = 36/99$.

Divide top and bottom by 9: $4/11$.

Example 2. Arrange $\frac{3}{4}$, $\frac{5}{7}$ and $\frac{7}{9}$ in ascending order.

Solution. $\frac{3}{4} = 0.75$, $\frac{5}{7} = 0.714$, $\frac{7}{9} = 0.778$.

Ascending order: $\frac{5}{7}$, $\frac{3}{4}$, $\frac{7}{9}$.

1.2.3 Squares, cubes & roots

What it is: Fast mental methods for squaring, cubing and taking roots, which save time in simplification questions.

How to identify it

- "Find the value of 47^2 ", " $\sqrt{7569}$ ", "cube root of 17576", or any simplification built on perfect squares and cubes.

How to solve it

- Square via a nearby round number: $47^2 = (47 + 3)(47 - 3) + 3^2 = 50 \times 44 + 9 = 2209$.
- Numbers ending in 5: for $n5$, the square is $n(n+1)$ followed by 25 ($65^2 = 4225$).
- Square root: squares end only in 0, 1, 4, 5, 6, 9. Use nearby perfect squares to get the tens digit, the last digit to choose between two candidates.
- Cube root of a perfect cube (up to 6 digits): the last digit of the cube fixes the last digit of the root (1→1, 2→8, 3→7, 4→4, 5→5, 6→6, 7→3, 8→2, 9→9). The leading digits, compared with 1, 8, 27, 64, 125, 216, 343, 512, 729, fix the tens digit.

Example 1. Find $\sqrt{7569}$.

Solution. $80^2 = 6400$ and $90^2 = 8100$, so the root is between 80 and 90.

The number ends in 9, so the root ends in 3 or 7.

$85^2 = 7225$ is less than 7569, so the root is above 85. The root is 87 ($87^2 = 7569$).

Example 2. Find the cube root of 17576.

Solution. The last digit is 6, so the root ends in 6.

Remove the last three digits to leave 17. Since $8 \leq 17 < 27$, the tens digit is 2.

Root = 26. Check: $26^3 = 676 \times 26 = 17576$.

1.2.4 Approximation

What it is: Replacing awkward numbers by nearby friendly ones to get an answer quickly when the options are well separated.

How to identify it

- The question says "approximately", shows $\approx$ or "?", and the numbers are long decimals.
- Options are far apart (not 1451 versus 1452).

How to solve it

- Round each number to one or two significant figures, keep the same operations, calculate, and pick the closest option.
- Use friendly fractions: $24.9\% \approx \frac{1}{4}$, $33.2\% \approx \frac{1}{3}$, $12.4\% \approx \frac{1}{8}$.
- Avoid rounding when you subtract two nearly equal numbers; the error becomes large.

Example 1. Estimate $49.8 \times 20.1 \div 9.9$.

Solution. Round to $50 \times 20 \div 10$.

$50 \times 20 = 1000$, and $1000 \div 10 = 100$. The answer is about 100.

Example 2. Estimate 39.97% of 2499 + 14.9% of 1199.

Solution. $39.97\% \approx 40\%$ and $2499 \approx 2500$, giving 1000.

$14.9\% \approx 15\%$ and $1199 \approx 1200$, giving 180.

Total ≈ 1180 .

1.3 Percentages

1.3.1 Basic percentage

What it is: Percent means "per hundred". $x\%$ of $N = (x/100) \times N$.

How to identify it

- "What percent of...", "x% of a number", "is what percent of", comparison of two quantities.
- Election and exam-marks questions that give percentages of a total.

How to solve it

- $x\%$ of $N = x \times N / 100$.
- A is what percent of $B = (A / B) \times 100$.
- Learn the fraction forms: $1/2 = 50\%$, $1/3 = 33.33\%$, $1/4 = 25\%$, $1/5 = 20\%$, $1/6 = 16.67\%$, $1/7 = 14.28\%$, $1/8 = 12.5\%$, $1/9 = 11.11\%$, $1/10 = 10\%$, $1/11 = 9.09\%$, $1/12 = 8.33\%$.
- $x\%$ of $y = y\%$ of x , so pick whichever is easier.

Example 1. Find 18% of 450, and say what percent 63 is of 350.

Solution. 18% of 450 = $0.18 \times 450 = 81$.

$63 \div 350 \times 100 = 18\%$.

Example 2. A candidate gets 45% of the votes and loses by 600 votes to the only other candidate. How many votes were cast?

Solution. The other candidate gets 55%. The gap is $55\% - 45\% = 10\%$.

10% of the total = 600, so the total = 6000.

1.3.2 Percentage change

What it is: How much a quantity has increased or decreased, expressed as a percentage of the original value.

How to identify it

- "Increased by x%", "what is the percentage increase/decrease", "A is x% more than B".
- Price rise and consumption cut questions.

How to solve it

- % change = $(\text{change} \div \text{original}) \times 100$. The original is always the denominator.
- New value = $\text{original} \times (100 \pm x)/100$.
- If A is x% more than B, then B is $100x/(100 + x)$ % less than A.
- To keep expenditure the same after a price rise of x%, consumption must fall by $100x/(100 + x)$ %.

Example 1. Sugar becomes 25% costlier. By what percent must a family cut its use to keep the spending the same?

Solution. Reduction = $100 \times 25 / (100 + 25) = 20\%$.

Example 2. A's salary is 20% more than B's. By what percent is B's salary less than A's?

Solution. Reduction = $100 \times 20 / (100 + 20) = 16.67\%$ (that is, $1/6$).

1.3.3 Successive percentages

What it is: Two or more percentage changes applied one after another. The net effect is not the simple sum of the changes.

How to identify it

- "Increased by 10% and then by 20%", "first increased, then decreased", repeated discounts or growth.
- Change in the area of a figure when its sides change by percentages.

How to solve it

- Net % change = $a + b + ab/100$, using + for an increase and – for a decrease.
- Or multiply the factors: +10% then +20% is $1.10 \times 1.20 = 1.32$, a 32% increase.
- For three changes, combine the first two, then combine the result with the third.

Example 1. A price rises by 20% and then falls by 20%. What is the net change?

Solution. Net = $20 - 20 + (20)(-20)/100 = -4\%$.

The price is 4% lower. (Check: $1.2 \times 0.8 = 0.96$.)

Example 2. The side of a square is increased by 10%. What happens to its area?

Solution. Area depends on the side twice: $10 + 10 + (10 \times 10)/100 = 21\%$.

The area increases by 21%.

1.3.4 Population & depreciation

What it is: Growth or decline at a fixed percent per period compounds, because each period works on the new value.

How to identify it

- "Population increases by x% per year", "value of a machine depreciates at x% per annum", "after n years", "n years ago".

How to solve it

- Growth: final = $P \times (1 + r/100)^n$.
- Depreciation: final = $P \times (1 - r/100)^n$.
- If the rate changes each year, multiply the separate factors.
- To find an earlier value, divide by the factor instead of multiplying.

Example 1. A town has 20,000 people and grows 10% per year. Population after 2 years?

Solution. $20000 \times 1.1 \times 1.1 = 24,200$.

Example 2. A machine costing ₹1,00,000 depreciates 10% every year. Its value after 3 years?

Solution. $1,00,000 \times 0.9^3 = 1,00,000 \times 0.729$.

Value = ₹72,900.

1.4 Ratio, Proportion & Variation

1.4.1 Ratio

What it is: A ratio compares two quantities of the same kind by division. The ratio a : b is the fraction a/b.

How to identify it

- "Divide ₹X in the ratio...", "A : B = 2 : 3 and B : C = 4 : 5, find A : B : C".
- "The ratio changes when the same amount is added to both".

How to solve it

- To divide N in the ratio a : b, the shares are $a/(a+b) \times N$ and $b/(a+b) \times N$.
- Write the quantities as ak and bk and use the conditions to find k.
- To join A : B and B : C, make the B terms equal using their LCM, then read off A : B : C.
- Compound ratio = product of the first terms : product of the second terms.

Example 1. If A : B = 2 : 3 and B : C = 4 : 5, find A : B : C.

Solution. B is 3 in the first ratio and 4 in the second. The LCM of 3 and 4 is 12.

A : B = 8 : 12 and B : C = 12 : 15.

So A : B : C = 8 : 12 : 15.

Example 2. Two numbers are in the ratio 3 : 5. When 10 is added to each, the ratio becomes 5 : 7. Find the numbers.

Solution. Let the numbers be 3k and 5k. Then $(3k + 10)/(5k + 10) = 5/7$.

$21k + 70 = 25k + 50$, so $4k = 20$ and $k = 5$.

The numbers are 15 and 25. Check: $25/35 = 5/7$.

1.4.2 Proportion

What it is: Proportion is the equality of two ratios: $a : b = c : d$. The product of the extremes equals the product of the means, so $ad = bc$.

How to identify it

- "Find the fourth proportional", "third proportional", "mean proportional between".
- "If 6 pens cost ₹90, find the cost of 10 pens" (a unitary-method proportion).

How to solve it

- Fourth proportional to $a, b, c = bc/a$.
- Third proportional to a and $b = b^2/a$.
- Mean proportional between a and $b = \sqrt{ab}$.
- For unitary problems, find the value of one unit, then multiply.

Example 1. Find the mean proportional between 4 and 36, and the third proportional to 6 and 12.

Solution. Mean proportional $= \sqrt{(4 \times 36)} = \sqrt{144} = 12$.

Third proportional to 6 and 12 $= 12^2/6 = 24$.

Example 2. If 6 pens cost ₹90, what do 10 pens cost?

Solution. One pen costs $90 \div 6 = ₹15$.

Ten pens cost $10 \times 15 = ₹150$.

1.4.3 Direct & inverse variation

What it is: Two quantities vary directly if they rise and fall together (y/x stays constant). They vary inversely if one rises as the other falls ($x \times y$ stays constant).

How to identify it

- Direct: cost and quantity, distance and time at a fixed speed, work done and number of workers.
- Inverse: men and days for the same job, speed and time for the same distance.
- Several linked quantities (men, hours per day, days) use the chain method.

How to solve it

- Direct: $x_1/y_1 = x_2/y_2$.
- Inverse: $x_1 \times y_1 = x_2 \times y_2$.
- Chain: men $\times$ hours $\times$ days is proportional to the work done. Compare the two situations.

Example 1. 12 men build a wall in 20 days. How many days will 15 men take?

Solution. More men means fewer days (inverse): $12 \times 20 = 15 \times d$.

$d = 240 \div 15 = 16$ days.

Example 2. 5 men working 8 hours a day make 400 items in 10 days. How many items do 8 men make working 5 hours a day for 12 days?

Solution. Man-hours in the first case $= 5 \times 8 \times 10 = 400$, which gives 400 items, so 1 item per man-hour.

Man-hours in the second case $= 8 \times 5 \times 12 = 480$.

Items $= 480$.

1.4.4 Componendo & dividendo

What it is: A shortcut: if $a/b = c/d$ then $(a + b)/(a - b) = (c + d)/(c - d)$.

How to identify it

- Equations such as $(x + y)/(x - y) = 7/3$ where you need $x : y$.
- Expressions where both a sum and a difference of the same two terms appear on top and bottom.

How to solve it

- Componendo: $(a + b)/b = (c + d)/d$. Dividendo: $(a - b)/b = (c - d)/d$.
- Combined: $(a + b)/(a - b) = (c + d)/(c - d)$.
- Read it backwards too: if $(P + Q)/(P - Q) = m/n$ then $P/Q = (m + n)/(m - n)$.

Example 1. If $(x + y)/(x - y) = 7/3$, find $x : y$.

Solution. Using the shortcut in reverse: $x/y = (7 + 3)/(7 - 3) = 10/4 = 5/2$.

Check: $x = 5, y = 2$ gives $7/3$.

Example 2. If $(3a + 2b)/(3a - 2b) = 5/3$, find $a : b$.

Solution. $3a/2b = (5 + 3)/(5 - 3) = 8/2 = 4$.

So $3a = 8b$ and $a/b = 8/3$.

Check: $a = 8, b = 3$ gives $(24 + 6)/(24 - 6) = 30/18 = 5/3$.

1.5 Averages

1.5.1 Simple average

What it is: The average (mean) is the sum of the values divided by how many there are.

How to identify it

- "Average of...", "mean", "average age of a class", "the average of five numbers is...".
- Missing value or wrong value found later.

How to solve it

- Average = Sum $\div$ Number of items, so Sum = Average $\times$ Number of items.
- For consecutive numbers or an arithmetic progression, average = (first + last)/2 = the middle term.
- Average of the first n natural numbers = $(n + 1)/2$. First n even numbers: $n + 1$. First n odd numbers: n .
- If every value rises by k , the average rises by k .

Example 1. The average of 5 numbers is 24. Four of them are 20, 22, 26 and 28. Find the fifth.

Solution. Total = $5 \times 24 = 120$. The four known numbers add to 96.

Fifth number = $120 - 96 = 24$.

Example 2. The average of 7 consecutive numbers is 20. What is the largest?

Solution. The middle (4th) number is the average, 20.

The numbers run 17, 18, 19, 20, 21, 22, 23. The largest is 23.

1.5.2 Weighted average

What it is: An average in which groups of different sizes count in proportion to their sizes.

How to identify it

- "Average of two sections combined", "average price of a mixture", "some items at one price, others at another".

How to solve it

- Weighted average = $(n_1 a_1 + n_2 a_2) \div (n_1 + n_2)$.
- The combined average always lies between the two group averages, nearer to the bigger group.
- Alligation shortcut: $n_1 : n_2 = (a_2 - \text{average}) : (\text{average} - a_1)$.

Example 1. Class A has 40 students with average 70, and class B has 60 students with average 80. Find the combined average.

Solution. Total marks = $40 \times 70 + 60 \times 80 = 2800 + 4800 = 7600$.

Average = $7600 \div 100 = 76$.

Example 2. Rice at ₹40/kg is mixed with rice at ₹55/kg in the ratio 2 : 3. What is the average cost per kg?

Solution. Cost of 5 kg = $2 \times 40 + 3 \times 55 = 80 + 165 = 245$.

Average = $245 \div 5 = ₹49/\text{kg}$.

1.5.3 Average speed

What it is: Average speed is total distance divided by total time. It is not the plain average of two speeds.

How to identify it

- "Goes at x km/h and returns at y km/h", "first half at..., second half at...", "average speed for the whole journey".

How to solve it

- Equal distances at speeds a and b : average speed = $2ab/(a + b)$.
- Equal times at speeds a and b : average speed = $(a + b)/2$.
- Unequal parts: total distance $\div$ total time.

Example 1. A car travels to a town at 60 km/h and returns at 40 km/h. Find its average speed.

Solution. The distances are equal, so average = $2 \times 60 \times 40 \div (60 + 40) = 4800 \div 100$.

Average speed = 48 km/h.

Example 2. A man travels for 2 hours at 30 km/h and then for 3 hours at 50 km/h. Find his average speed.

Solution. Distance = $2 \times 30 + 3 \times 50 = 60 + 150 = 210$ km.

Time = 5 hours, so average speed = $210 \div 5 = 42$ km/h.

1.5.4 Replacement & addition problems

What it is: Working out how the average of a group changes when a member joins, leaves or is replaced, or finding the member's value from the change.

How to identify it

- "A new person joins and the average rises by...", "replaced by", "one student left".

How to solve it

- Replacement: value of new member – value of old member = (change in average) $\times$ (group size).
- Joining: new member's value = old average + (change in average) $\times$ (new group size).
- Leaving or any other case: write the total before and after; the difference is the member's value.

Example 1. The average weight of 8 men increases by 2.5 kg when a new man replaces one who weighs 60 kg. Find the weight of the new man.

Solution. Total weight rises by $8 \times 2.5 = 20$ kg.

New man = $60 + 20 = 80$ kg.

Example 2. The average age of 10 people is 30. When an 11th person joins, the average becomes 31. Find the new person's age.

Solution. Old total = 300. New total = $11 \times 31 = 341$.

New person = $341 - 300 = 41$ years.

1.6 Problems on Ages

1.6.1 Present, past & future ages

What it is: Word problems that link people's ages at different times. Everyone's age changes by the same number of years, so the difference between two ages never changes.

How to identify it

- "x years ago", "after n years", "twice as old as", "sum of their ages".

How to solve it

- Let the youngest (or the one you are asked about) be x.
- Write each sentence as an equation. For a time shift of t years, add t (future) or subtract t (past) to every person's age.
- Solve the linear equation and check against every sentence.

Example 1. A father is 3 times as old as his son. After 12 years he will be twice as old. Find their present ages.

Solution. Son = x, father = 3x.

$3x + 12 = 2(x + 12)$, so $3x + 12 = 2x + 24$ and $x = 12$.

Son is 12 and father is 36.

Example 2. Ravi is 5 years older than Mohan. Four years ago Ravi was twice as old as Mohan. Find their ages now.

Solution. Mohan = x, Ravi = $x + 5$.

$(x + 5) - 4 = 2(x - 4)$, so $x + 1 = 2x - 8$ and $x = 9$.

Mohan is 9 and Ravi is 14. Check: four years ago they were 5 and 10.

1.6.2 Ratio-based age problems

What it is: Ages are given as ratios, sometimes at two different times.

How to identify it

- "The ratio of their ages is 5 : 7", "after 8 years the ratio will be..."

How to solve it

- Present ages = ak and bk .
- Apply the time shift to each age, then use the second ratio or the stated sum or difference to find k .
- The difference $(b - a)k$ stays the same at all times, which often gives a shortcut.

Example 1. The ages of A and B are in the ratio 5 : 7. Four years ago the sum of their ages was 52. Find their present ages.

Solution. Present ages $5k$ and $7k$. Four years ago the sum was $12k - 8 = 52$.

$12k = 60$, so $k = 5$.

Ages are 25 and 35. Check: four years ago $21 + 31 = 52$.

Example 2. The ages of a mother and daughter are in the ratio 7 : 2. After 8 years the ratio will be 5 : 2. Find their present ages.

Solution. Mother $7k$, daughter $2k$. $(7k + 8)/(2k + 8) = 5/2$.

$14k + 16 = 10k + 40$, so $4k = 24$ and $k = 6$.

Mother is 42 and daughter is 12. After 8 years they will be 50 and 20, which is 5 : 2.

1.7 Partnership

1.7.1 Simple partnership

What it is: Partners invest money for the same period and share the profit or loss in the ratio of their investments.

How to identify it

- "A and B start a business with ₹... and ₹...", "share the profit", all invested for the whole year.

How to solve it

- Profit ratio = ratio of investments.
- Each partner's share = (their parts ÷ total parts) × total profit.
- If one share and the ratio are known, scale up to the total.

Example 1. A invests ₹30,000 and B invests ₹50,000. The profit is ₹24,000. Find each share.

Solution. Ratio = 3 : 5, total 8 parts, 1 part = ₹3,000.

A gets $3 \times 3000 = ₹9,000$ and B gets $5 \times 3000 = ₹15,000$.

Example 2. Three partners invest ₹20,000, ₹30,000 and ₹50,000. C's share of the profit is ₹15,000. Find the total profit.

Solution. Ratio = 2 : 3 : 5, total 10 parts. C holds 5 parts = ₹15,000.

One part = ₹3,000, so total profit = $10 \times 3000 = ₹30,000$.

1.7.2 Compound partnership

What it is: Partners invest different amounts for different lengths of time, or change their capital. The profit is shared in the ratio of (capital × time).

How to identify it

- "B joins after 4 months", "A withdraws ₹... at the end of 6 months", different time periods.

How to solve it

- Profit ratio = $\text{capital}_1 \times \text{months}_1 : \text{capital}_2 \times \text{months}_2 : \dots$
- If a partner's capital changes, add up capital × months for each separate period.
- Then divide the profit in that ratio.

Example 1. A invests ₹6,000 for the whole year and B invests ₹8,000 for 9 months. In what ratio is the profit shared?

Solution. A: $6000 \times 12 = 72,000$. B: $8000 \times 9 = 72,000$.

Ratio = 1 : 1.

Example 2. A starts a business with ₹5,000. After 4 months B joins with ₹6,000. The profit at the end of the year is ₹5,400. Find each share.

Solution. A: $5000 \times 12 = 60,000$. B: $6000 \times 8 = 48,000$. Ratio = 5 : 4.
One part = $5400 \div 9 = ₹600$. A gets ₹3,000 and B gets ₹2,400.

1.7.3 Working vs. sleeping partner

What it is: A working partner manages the business and is paid a commission or salary before the profit is shared. A sleeping partner only invests money.

How to identify it

- "A is a working partner and gets 10% of the profit for managing", "B is a sleeping partner", "salary".

How to solve it

- Deduct the commission or salary from the total profit first.
- Divide the remaining profit in the ratio of capital $\times$ time.
- The working partner's total = commission (or salary) + investment share.

Example 1. A (working) and B (sleeping) invest ₹40,000 and ₹60,000. A receives 10% of the profit for managing. The profit is ₹20,000. Find A's total.

Solution. Commission = 10% of 20,000 = ₹2,000. Remaining = ₹18,000, shared 2 : 3.
A's share = 7,200 and B's share = 10,800.
A's total = 2,000 + 7,200 = ₹9,200.

Example 2. A manages the business for a salary of ₹1,000 a month and invests ₹20,000. B invests ₹30,000. The yearly profit before salary is ₹32,000. Find what each receives.

Solution. Annual salary = $12 \times 1000 = ₹12,000$. Remaining profit = ₹20,000, shared 2 : 3.
A's share = 8,000 and B's share = 12,000.
A receives 12,000 + 8,000 = ₹20,000 in all, and B receives ₹12,000.

1.8 Profit & Loss

1.8.1 Cost price, selling price, profit & loss

What it is: Cost price (CP) is what the seller pays; selling price (SP) is what the buyer pays. Profit = SP – CP, Loss = CP – SP.

How to identify it

- "Gain %", "loss %", "sold at a profit of", "find the cost price".

How to solve it

- Profit % = $(\text{Profit} \div \text{CP}) \times 100$ and Loss % = $(\text{Loss} \div \text{CP}) \times 100$. Percentages are on CP unless the question says otherwise.
- $\text{SP} = \text{CP} \times (100 + P)/100$ for a profit, $\text{CP} \times (100 - L)/100$ for a loss.
- $\text{CP} = \text{SP} \times 100/(100 + P)$ or $\text{SP} \times 100/(100 - L)$.
- If two articles are sold at the same price, one at a gain of p% and the other at a loss of p%, there is always a loss of $p^2/100$ %.

Example 1. An article bought for ₹800 is sold for ₹920. Find the profit percent.

Solution. Profit = $920 - 800 = ₹120$.
Profit % = $120 \div 800 \times 100 = 15\%$.

Example 2. A shopkeeper gains 5% by selling a bag for ₹1,260. At what price must he sell it to gain 8%?

Solution. CP = $1260 \times 100/105 = ₹1,200$.
SP at 8% gain = $1200 \times 1.08 = ₹1,296$.

1.8.2 Discount & marked price

What it is: The marked price (MP) is the price printed on the article. A discount is a reduction on the marked price, so $SP = MP - \text{discount}$.

How to identify it

- "Marked at", "list price", "discount of x%", "after allowing a discount", "marked up by".

How to solve it

- Discount % is always on MP, while profit % is on CP.
- $SP = MP \times (100 - d)/100$.
- $MP = CP \times (100 + P)/(100 - d)$, which links marked price, profit and discount.
- Start with $CP = 100$ when percentages are given and no amounts.

Example 1. A shopkeeper marks a watch 40% above cost and allows a 25% discount. Find his profit percent.

Solution. Take $CP = 100$. $MP = 140$. $SP = 140 \times 0.75 = 105$.

Profit = 5%.

Example 2. A dealer's cost is ₹45,000. What should he mark the item at so that, after a 10% discount, he still gains 15%?

Solution. $MP = 45000 \times 1.15 \div 0.9 = 51,750 \div 0.9$.

$MP = ₹57,500$.

1.8.3 Successive discounts

What it is: Two or more discounts allowed one after another. Together they equal one smaller discount than the simple sum.

How to identify it

- "Successive discounts of 20% and 10%", "discount series", "equivalent single discount".

How to solve it

- Net discount % = $a + b - ab/100$.
- Or multiply the factors: 20% and 10% give $0.8 \times 0.9 = 0.72$, so the net discount is 28%.
- The order of the discounts does not change the final price.

Example 1. Find the single discount equal to successive discounts of 20% and 10%.

Solution. Net = $20 + 10 - (20 \times 10)/100 = 30 - 2$.

Single discount = 28%.

Example 2. A TV is marked ₹50,000 and is sold after successive discounts of 10% and 20%. Find the price paid.

Solution. $50,000 \times 0.9 \times 0.8 = 50,000 \times 0.72$.

Price = ₹36,000.

1.8.4 Dishonest dealings

What it is: A seller makes extra profit by using a false weight or measure, either alone or on top of a markup.

How to identify it

- "Uses a weight of 900 g for 1 kg", "sells at cost price but uses a false weight", "claims to sell at cost".

How to solve it

- If sold at cost price: Gain % = $(\text{true weight} - \text{false weight}) \div \text{false weight} \times 100$.
- Equivalent factor = $\text{true weight} \div \text{false weight}$.
- If there is also a markup, multiply the factors: total factor = $(\text{true} \div \text{false}) \times (1 + \text{markup})$. Gain % = factor - 1 (as a percent).

Example 1. A shopkeeper sells at cost price but uses a 900 g weight for 1 kg. Find his gain percent.

Solution. He gives 900 g but charges for 1000 g. Gain = $100 \div 900 \times 100$.

Gain % = 11.11%.

Example 2. A trader uses an 800 g weight for 1 kg and also sells at 20% above cost. Find his total gain percent.

Solution. False-weight factor = $1000 \div 800 = 1.25$. Markup factor = 1.2.

Total factor = $1.25 \times 1.2 = 1.5$, so the gain is 50%.

1.9 Simple & Compound Interest

1.9.1 Simple interest

What it is: Interest calculated only on the original sum (principal), at the same amount every year.

How to identify it

- "Simple interest", "per annum" with no mention of compounding, "in how many years will it double or triple".

How to solve it

- $SI = P \times R \times T \div 100$, and $\text{Amount} = P + SI$.
- Convert months to years (9 months = 0.75 year).
- A sum doubles when $R \times T = 100$; triples when $R \times T = 200$.

Example 1. Find the simple interest on ₹8,000 at 7.5% per annum for 4 years.

Solution. $SI = 8000 \times 7.5 \times 4 \div 100 = 8000 \times 30 \div 100$.

$SI = ₹2,400$.

Example 2. At what simple interest rate will a sum become three times itself in 10 years?

Solution. Interest needed = $2P$, so $2P = P \times R \times 10 \div 100$.

$R = 20\%$ per annum.

1.9.2 Compound interest

What it is: Interest earned on the principal plus the interest already added in earlier periods.

How to identify it

- "Compound interest", "compounded annually, half-yearly or quarterly".

How to solve it

- $\text{Amount} = P \times (1 + R/100)^n$, and $CI = \text{Amount} - P$.
- Half-yearly compounding: use rate $R/2$ and $2n$ periods. Quarterly: $R/4$ and $4n$ periods.
- If the rate differs each year, multiply $(1 + r/100)$ for each year.

Example 1. Find the compound interest on ₹10,000 at 10% per annum for 2 years.

Solution. $\text{Amount} = 10000 \times 1.1 \times 1.1 = 12,100$.

$CI = 12,100 - 10,000 = ₹2,100$.

Example 2. Find the amount on ₹16,000 at 20% per annum, compounded half-yearly, for 1 year.

Solution. Rate per half-year = 10%, number of periods = 2.

$\text{Amount} = 16000 \times 1.1 \times 1.1 = ₹19,360$.

1.9.3 Difference between CI and SI

What it is: For 2 and 3 years the gap between compound and simple interest on the same sum has a ready formula.

How to identify it

- "The difference between the compound and simple interest is ₹...", "find the sum".

How to solve it

- 2 years: $CI - SI = P \times (R/100)^2$.
- 3 years: $CI - SI = P \times (R/100)^2 \times (3 + R/100)$.
- Use the formula backwards to find P or R .

Example 1. The difference between CI and SI on a sum for 2 years at 5% per annum is ₹75. Find the sum.

Solution. $75 = P \times (0.05)^2 = P \times 0.0025$.

$P = 75 \div 0.0025 = ₹30,000$.

Example 2. Find the difference between CI and SI on ₹10,000 at 10% per annum for 3 years.

Solution. Difference = $10000 \times (0.1)^2 \times (3 + 0.1) = 10000 \times 0.01 \times 3.1 = ₹310$.

Check: $SI = 3,000$, $CI = 10000 \times (1.331 - 1) = 3,310$.

1.9.4 Installments

What it is: A loan or purchase price paid back in equal parts at regular intervals. Each installment covers the interest and part of the principal.

How to identify it

- "Payable in two equal annual installments", "cleared in equal yearly installments".

How to solve it

- Compound interest: the loan equals the present value of the installments. For 2 installments x at rate r : $\text{Loan} = x/(1+r) + x/(1+r)^2$.
- Simple interest: the installments paid early earn interest for the time left. For two yearly installments: $x \times (1+r) + x = \text{Loan} \times (1+2r)$.
- Find x from the equation.

Example 1. A loan of ₹21,000 is repaid in two equal annual installments at 10% compound interest. Find the installment.

Solution. $21000 = x/1.1 + x/1.21 = x \times (1.1 + 1)/1.21 = x \times 2.1/1.21$.
 $x = 21000 \times 1.21 \div 2.1 = ₹12,100$.

Example 2. A loan of ₹3,500 at 10% simple interest is cleared in two equal yearly installments. Find the installment.

Solution. Amount owed after 2 years = $3500 \times 1.2 = 4,200$.
The first installment earns one year of interest: $1.1x + x = 4200$, so $2.1x = 4200$.
 $x = ₹2,000$.

1.9.5 Effective rate of interest

What it is: The actual yearly rate you earn when interest is compounded more than once a year.

How to identify it

- "Effective annual rate", "equivalent annual rate", "compounded half-yearly or quarterly, what yearly rate is this the same as".

How to solve it

- Compounded k times a year at $R\%$: effective rate = $[(1 + R/(100k))^k - 1] \times 100$.
- Half-yearly: $k = 2$. Quarterly: $k = 4$. Monthly: $k = 12$.
- The effective rate is always higher than the stated rate.

Example 1. Find the effective annual rate for 10% per annum compounded half-yearly.

Solution. Growth factor = $(1 + 0.05)^2 = 1.1025$.
Effective rate = 10.25%.

Example 2. Find the effective annual rate for 8% per annum compounded quarterly.

Solution. Growth factor = $(1.02)^4 = 1.0824$ (since $1.02^2 = 1.0404$ and $1.0404^2 \approx 1.0824$).
Effective rate $\approx 8.24\%$.

1.10 Mixtures & Alligations

1.10.1 Alligation rule

What it is: A quick way to find the ratio in which two ingredients of different price or strength must be mixed to get a required mean value.

How to identify it

- "In what ratio must ... be mixed to get ...", "mean price", "two varieties", percentage strength of solutions.

How to solve it

- Let the cheaper (lower) value be c , the dearer (higher) value d and the mean m .
- Quantity of cheaper : quantity of dearer = $(d - m) : (m - c)$.
- The mean must lie between c and d . Cross-subtract along the diagonals and read the ratio.

Example 1. In what ratio must rice at ₹36/kg be mixed with rice at ₹42/kg to get a mixture worth ₹39.50/kg?

Solution. Cheaper : dearer = $(42 - 39.5) : (39.5 - 36) = 2.5 : 3.5$.
Ratio = 5 : 7.
Check: $(5 \times 36 + 7 \times 42) \div 12 = 474 \div 12 = 39.5$.

Example 2. In what ratio must a 30% alcohol solution be mixed with a 50% solution to get a 45% solution?

Solution. Weaker : stronger = $(50 - 45) : (45 - 30) = 5 : 15$.

Ratio = 1 : 3.

1.10.2 Replacement of a mixture

What it is: Repeatedly taking out part of a liquid and replacing it with another liquid (usually water) dilutes the original liquid by a fixed fraction each time.

How to identify it

- "A vessel contains x litres of milk; y litres are removed and replaced with water; this is repeated n times".

How to solve it

- Original liquid left after n operations = $x \times (1 - y/x)^n$.
- Fraction of the original in the vessel = $(1 - y/x)^n$.
- Water present = total – original liquid left.

Example 1. A 40-litre vessel is full of pure milk. 4 litres are removed and replaced with water, and the process is repeated once more. How much milk is left?

Solution. Fraction left after each step = $1 - 4/40 = 0.9$.

Milk left = $40 \times 0.9 \times 0.9 = 32.4$ litres.

Example 2. A container has 64 litres of milk. 16 litres are taken out and replaced with water, three times. Find the milk left and the milk : water ratio.

Solution. Fraction left each time = $1 - 16/64 = 3/4$.

Milk left = $64 \times (3/4)^3 = 64 \times 27/64 = 27$ litres.

Water = $64 - 27 = 37$ litres, so milk : water = 27 : 37.

1.11 Time & Work

1.11.1 Work-rate

What it is: If a person completes a job in n days, their one-day work is $1/n$ of the job. Rates of people working together add up.

How to identify it

- "A can do a job in 10 days and B in 15 days; together they take...".
- "A and B together finish in 8 days; A alone takes 12; find B alone".

How to solve it

- Rate of A = $1/a$ per day. Together: $1/a + 1/b = (a + b)/(ab)$, so days together = $ab/(a + b)$.
- Work done in d days = $d \times \text{rate}$.
- To find one worker from the pair, subtract: $1/b = 1/t - 1/a$.
- If someone leaves or joins midway, calculate the work done in each stage and add.

Example 1. A can do a job in 10 days and B in 15 days. How long will they take together?

Solution. $1/10 + 1/15 = 3/30 + 2/30 = 5/30 = 1/6$ per day.

They take 6 days.

Example 2. A can finish a job in 12 days. A and B together finish it in 8 days. How long does B take alone?

Solution. B's rate = $1/8 - 1/12 = 3/24 - 2/24 = 1/24$.

B alone takes 24 days.

1.11.2 Efficiency method

What it is: Take the total work as the LCM of the given days. Then each worker's efficiency (units of work per day) is a whole number, and the arithmetic becomes simple.

How to identify it

- Three or more workers with different times, workers leaving or joining, "work on alternate days".

How to solve it

- Total work = LCM of the days taken by the individuals.
- Efficiency = total work $\div$ days taken.
- Add efficiencies for workers together. Time = total work $\div$ combined efficiency.
- For alternate days, compute the work in one full cycle, count whole cycles, then handle the leftover.

Example 1. A, B and C can finish a job in 20, 30 and 60 days. How long will they take together?

Solution. Total work = LCM(20, 30, 60) = 60 units.

Efficiencies: A = 3, B = 2, C = 1 per day, together 6 per day.

Time = $60 \div 6 = 10$ days.

Example 2. A can finish a job in 12 days and B in 18 days. They work on alternate days, starting with A. How long does the job take?

Solution. Total work = LCM(12, 18) = 36 units. A does 3 per day, B does 2 per day.

Two-day cycle = 5 units. After 14 days (7 cycles) the work done is 35 units.

One unit is left. The next day belongs to A, who needs $\frac{1}{3}$ of a day. Total = $14 \frac{1}{3}$ days.

1.11.3 Work & wages

What it is: Payment for a job is shared in proportion to the work each person has done.

How to identify it

- "Total wages ₹...", "they work together and are paid", "share of each".

How to solve it

- Wage ratio = ratio of work done = efficiency $\times$ time.
- If they work for the same time, the ratio equals the ratio of efficiencies, which is the inverse of the ratio of their individual days.
- Find each person's share of the work first, then split the money in that ratio.

Example 1. A can finish a job in 6 days and B in 12 days. They work together on a job that pays ₹900. Find each share.

Solution. Efficiency ratio = $\frac{1}{6} : \frac{1}{12} = 2 : 1$.

A gets $\frac{2}{3} \times 900 = ₹600$ and B gets ₹300.

Example 2. A and B can finish a job in 12 and 18 days. With C's help it is finished in 4 days and the wage is ₹1,800. Find each person's share.

Solution. Rates: $\frac{1}{12} + \frac{1}{18} = \frac{5}{36}$. All three together: $\frac{1}{4} = \frac{9}{36}$, so C = $\frac{4}{36} = \frac{1}{9}$ (9 days alone).

Work done in 4 days: A = $\frac{12}{36}$, B = $\frac{8}{36}$, C = $\frac{16}{36}$. Ratio = 3 : 2 : 4 (9 parts, ₹200 each).

A gets ₹600, B gets ₹400 and C gets ₹800.

1.11.4 Pipes & cisterns

What it is: Work-rate applied to tanks. An inlet pipe does positive work (fills), an outlet pipe or leak does negative work (empties).

How to identify it

- "A pipe can fill a tank in...", "an outlet empties it in...", "a leak", "all three pipes open together".

How to solve it

- Inlet rate = $+\frac{1}{t}$ per hour, outlet rate = $-\frac{1}{t}$ per hour.
- Net rate = sum of all rates. Time to fill = $1 \div$ net rate.
- A leak: if a pipe fills a tank in x hours but takes y hours ($y > x$) because of a leak, the leak alone empties a full tank in $\frac{xy}{(y - x)}$ hours.

Example 1. Pipe A fills a tank in 6 hours and pipe B in 8 hours. Pipe C empties it in 12 hours. If all are open, how long to fill the tank?

Solution. Net rate = $\frac{1}{6} + \frac{1}{8} - \frac{1}{12} = \frac{4}{24} + \frac{3}{24} - \frac{2}{24} = \frac{5}{24}$.
Time = $24/5 = 4.8$ hours.

Example 2. A tank fills in 5 hours. Because of a leak it takes 6 hours. How long does the leak take to empty a full tank?

Solution. Time = $5 \times 6 \div (6 - 5)$.
The leak empties it in 30 hours.

1.12 Time, Speed & Distance

1.12.1 Basic formula

What it is: Distance = Speed $\times$ Time. All other problems in this topic are variations of this relation.

How to identify it

- "How far", "how long", "at what speed", "reaches late or early".

How to solve it

- Keep units consistent. km/h to m/s: multiply by $5/18$. m/s to km/h: multiply by $18/5$.
- For a fixed distance, speed and time are inversely proportional: speeds in ratio $a : b$ mean times in ratio $b : a$.
- Late/early problems: difference in times = $d/s_1 - d/s_2$.

Example 1. Convert 72 km/h into m/s and find the time to cover 500 m at this speed.

Solution. $72 \times 5/18 = 20$ m/s.

Time = $500 \div 20 = 25$ seconds.

Example 2. A man walking at 5 km/h reaches the office 20 minutes later than when he walks at 6 km/h. Find the distance.

Solution. Time difference = 20 minutes = $1/3$ hour.

$d/5 - d/6 = 1/3$, so $d/30 = 1/3$ and $d = 10$ km.

1.12.2 Relative speed

What it is: The speed of one moving object as seen from another. Speeds add when objects move towards or away from each other, and subtract when they move in the same direction.

How to identify it

- "Start from two places towards each other and meet", "overtakes", "chases", "catch up".

How to solve it

- Opposite directions: relative speed = $S_1 + S_2$.
- Same direction: relative speed = $S_1 - S_2$ (faster minus slower).
- Time to meet or catch up = gap (distance) $\div$ relative speed.

Example 1. Two towns are 360 km apart. Two cars leave them at the same time, at 50 km/h and 70 km/h, towards each other. When do they meet?

Solution. Relative speed = $50 + 70 = 120$ km/h.

Time = $360 \div 120 = 3$ hours.

Example 2. A thief runs at 8 km/h. A policeman 100 m behind chases him at 10 km/h. How long to catch him?

Solution. Relative speed = $10 - 8 = 2$ km/h. Gap = 0.1 km.

Time = $0.1 \div 2 = 0.05$ hour = 3 minutes.

1.12.3 Trains

What it is: Problems in which the length of the train matters. The train must cover its own length (and the length of whatever it crosses).

How to identify it

- "Crosses a pole, a man, a platform, a bridge, or another train".

How to solve it

- Pole or standing man: distance = length of train.
- Platform or bridge: distance = length of train + length of platform.
- Two trains crossing: distance = sum of both lengths; speed = sum of speeds (opposite) or difference (same direction).
- Convert km/h to m/s ($\times 5/18$) when lengths are in metres.

Example 1. A 150 m train at 54 km/h crosses a 250 m platform. How long does it take?

Solution. Speed = $54 \times 5/18 = 15$ m/s. Distance = $150 + 250 = 400$ m.
Time = $400 \div 15 = 26 \frac{2}{3}$ seconds.

Example 2. Trains 120 m and 180 m long run at 60 km/h and 40 km/h in opposite directions. How long do they take to cross each other?

Solution. Relative speed = 100 km/h = $100 \times 5/18 = 250/9$ m/s. Distance = 300 m.
Time = $300 \times 9 \div 250 = 10.8$ seconds.

1.12.4 Boats & streams

What it is: A boat's speed changes with the current of the stream. Moving with the stream (downstream) it is faster; against it (upstream) it is slower.

How to identify it

- "Boat", "stream", "upstream", "downstream", "speed in still water".

How to solve it

- Downstream speed $D = b + s$. Upstream speed $U = b - s$ (b = boat in still water, s = stream).
- Boat speed $b = (D + U)/2$. Stream speed $s = (D - U)/2$.
- Total time for a round trip over distance $d = d/(b + s) + d/(b - s)$.

Example 1. A boat goes 24 km downstream in 2 hours and returns in 3 hours. Find the speed of the boat and of the stream.

Solution. Downstream speed = $24 \div 2 = 12$ km/h. Upstream speed = $24 \div 3 = 8$ km/h.
Boat = $(12 + 8)/2 = 10$ km/h. Stream = $(12 - 8)/2 = 2$ km/h.

Example 2. A boat's speed in still water is 15 km/h and the stream flows at 3 km/h. How long is a round trip to a place 36 km away?

Solution. Downstream: $36 \div 18 = 2$ hours. Upstream: $36 \div 12 = 3$ hours.
Total = 5 hours.

1.12.5 Races & games

What it is: Races where one runner beats another by a distance or a time, or gives a head start.

How to identify it

- "A beats B by 20 m", "gives a start of", "dead heat", "wins by 10 seconds".

How to solve it

- If A beats B by x metres in a race of D metres, then when A has run D , B has run $D - x$. So speeds are in the ratio $D : (D - x)$.
- Beating by time t : B would need t more seconds to finish.
- Chain the races: use the same moment of time and compare distances.

Example 1. In a 100 m race A beats B by 10 m, and B beats C by 10 m. By how much does A beat C?

Solution. When A runs 100 m, B runs 90 m. When B runs 100 m, C runs 90 m.
So when B runs 90 m, C runs $90 \times 0.9 = 81$ m.
A beats C by $100 - 81 = 19$ m.

Example 2. A runs 1 km in 3 minutes and B in 3 minutes 20 seconds. By how much does A beat B?

Solution. A takes 180 s. B takes 200 s for 1000 m, so in 180 s B covers $1000 \times 180/200 = 900$ m.
A beats B by 100 m (or by 20 seconds).

1.12.6 Circular tracks

What it is: Runners move round a closed loop. When they meet depends on their relative speed and the length of the track.

How to identify it

- "Circular track", "meet for the first time", "meet at the starting point", "round the track".

How to solve it

- Opposite directions: first meeting after $L \div (S_1 + S_2)$.
- Same direction: first meeting after $L \div (S_1 - S_2)$.
- To meet again at the starting point: take the LCM of the times each takes to complete one round.

Example 1. Two men start together from the same point of a 400 m track, running at 6 m/s and 4 m/s. When do they first meet if they run (a) in the same direction (b) in opposite directions?

Solution. (a) Relative speed $6 - 4 = 2$ m/s, so time $= 400 \div 2 = 200$ seconds.

(b) Relative speed $6 + 4 = 10$ m/s, so time $= 400 \div 10 = 40$ seconds.

Example 2. A and B run round a track, taking 12 minutes and 18 minutes per round. When will they next be together at the starting point?

Solution. $\text{LCM}(12, 18) = 36$.

They meet at the start after 36 minutes.

1.12.7 Escalators

What it is: A person walks on a moving staircase. The person's own speed and the escalator's speed combine, so the steps counted differ from the visible steps.

How to identify it

- "Escalator", "moving staircase", "number of steps", "walks up (or down) the escalator".

How to solve it

- With the escalator: effective speed = person + escalator. Against it: person – escalator.
- Visible steps $N = \text{effective speed} \times \text{time}$. Steps the person counts = own speed $\times$ time.
- If steps counted in two trips are given, use that time is proportional to the steps counted (same walking speed), and form two equations for N .

Example 1. A man counts 15 steps walking down a moving escalator and 60 steps walking up against it, at the same walking speed. Find the number of visible steps.

Solution. Let the escalator carry e steps during the downward trip. Then $N = 15 + e$.

Walking up takes 4 times as long (60 vs 15 steps), so the escalator moves $4e$ steps against him: $N = 60 - 4e$.

$15 + e = 60 - 4e$ gives $e = 9$, so $N = 24$ steps.

Example 2. An escalator has 100 visible steps and moves up at 2 steps per second. A person climbs at 3 steps per second relative to the steps. Find the time taken and the steps the person counts.

Solution. Effective speed $= 3 + 2 = 5$ steps per second.

Time $= 100 \div 5 = 20$ seconds. Steps counted $= 3 \times 20 = 60$.

1.13 Algebra

1.13.1 Linear equations

What it is: Equations in which the variable has power 1. They may have one variable or several (simultaneous equations).

How to identify it

- "Find x", "the sum of two numbers is... and their difference is...", "3 pens and 2 pencils cost...".

How to solve it

- One variable: collect the x terms on one side, numbers on the other.
- Two variables: elimination (match coefficients, add or subtract) or substitution.
- For $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$: unique solution if $a_1/a_2 \neq b_1/b_2$; no solution if $a_1/a_2 = b_1/b_2 \neq c_1/c_2$; infinitely many if all three ratios are equal.

Example 1. 3 pens and 2 pencils cost ₹36. 2 pens and 3 pencils cost ₹34. Find the price of a pen and a pencil.

Solution. $3p + 2c = 36$ and $2p + 3c = 34$.

Multiply the first by 3 and the second by 2: $9p + 6c = 108$ and $4p + 6c = 68$.

Subtract: $5p = 40$, so $p = 8$. Then $24 + 2c = 36$, so $c = 6$.

Pen ₹8 and pencil ₹6. Check: $16 + 18 = 34$.

Example 2. Solve $(2x + 3)/5 = (x - 1)/2 + 1$.

Solution. Multiply by 10: $2(2x + 3) = 5(x - 1) + 10$.

$4x + 6 = 5x + 5$, so $x = 1$.

Check: left = $5/5 = 1$, right = $0 + 1 = 1$.

1.13.2 Quadratic equations

What it is: Equations of the form $ax^2 + bx + c = 0$ ($a \neq 0$). They have two roots.

How to identify it

- An x^2 term, "find the roots", "nature of the roots", "sum and product of roots".
- "Compare x and y" questions where both come from quadratics.

How to solve it

- Factorise: find two numbers with product ac and sum b , split the middle term.
- Formula: $x = [-b \pm \sqrt{b^2 - 4ac}] / 2a$.
- Sum of roots = $-b/a$. Product of roots = c/a .
- Discriminant $D = b^2 - 4ac$: $D > 0$ two real roots, $D = 0$ equal roots, $D < 0$ no real roots.

Example 1. Solve $x^2 - 7x + 12 = 0$.

Solution. Find two numbers with product 12 and sum 7: 3 and 4.

$(x - 3)(x - 4) = 0$, so $x = 3$ or $x = 4$.

Example 2. For $2x^2 - 5x + 3 = 0$ find the discriminant, and the sum and product of the roots.

Solution. $D = 25 - 24 = 1$, which is positive, so the roots are real and different.

Sum = $5/2$ and product = $3/2$.

The roots are $(5 \pm 1)/4$, that is $3/2$ and 1 .

1.13.3 Inequalities

What it is: Statements using $<$, $>$, $\leq$ or $\geq$. Their solution is a range of values, not a single number.

How to identify it

- "Find the range of x", "how many integers satisfy", "solve the inequality".

How to solve it

- Solve as for an equation, but reverse the inequality sign when you multiply or divide by a negative number.
- Quadratic inequality: factorise, mark the roots on a number line, and test the regions.
- $|x| < a$ means $-a < x < a$. $|x| > a$ means $x < -a$ or $x > a$.
- When counting integers, check whether each end is included ($\leq$) or excluded ($<$).

Example 1. Solve $5 - 2x > 1$.

Solution. Subtract 5: $-2x > -4$.

Divide by -2 and reverse the sign: $x < 2$.

Example 2. How many integers satisfy $x^2 - 3x - 4 \leq 0$?

Solution. Factorise: $(x - 4)(x + 1) \leq 0$. The product is non-positive between the roots: $-1 \leq x \leq 4$.
Integers: $-1, 0, 1, 2, 3, 4$, which is 6 integers.

1.13.4 Progressions (AP, GP, HP)

What it is: An arithmetic progression (AP) adds a constant difference, a geometric progression (GP) multiplies by a constant ratio, and in a harmonic progression (HP) the reciprocals form an AP.

How to identify it

- "n-th term", "sum of first n terms", "common difference", "common ratio", "three numbers in AP/GP".

How to solve it

- AP: $a_n = a + (n - 1)d$. $S_n = n/2 \times [2a + (n - 1)d] = n/2 \times (\text{first} + \text{last})$.
- GP: $a_n = a \times r^{n-1}$. $S_n = a(r^n - 1)/(r - 1)$ for $r > 1$. Infinite GP with $|r| < 1$: $S = a/(1 - r)$.
- HP: take reciprocals, solve as an AP, take reciprocals back. Harmonic mean of a and b = $2ab/(a + b)$.
- Three numbers in AP: $a - d, a, a + d$. In GP: $a/r, a, ar$.

Example 1. Find the sum of the first 20 terms of the AP 3, 7, 11, ...

Solution. $a = 3$ and $d = 4$.

$$S = 20/2 \times [2 \times 3 + 19 \times 4] = 10 \times (6 + 76) = 820.$$

Example 2. For the GP 2, 6, 18, ... find the 6th term and the sum of the first 6 terms.

Solution. $a = 2$ and $r = 3$.

$$\text{6th term} = 2 \times 3^5 = 2 \times 243 = 486.$$

$$\text{Sum} = 2 \times (3^6 - 1)/(3 - 1) = 729 - 1 = 728.$$

1.13.5 Functions

What it is: A function f is a rule that gives exactly one output $f(x)$ for each allowed input x .

How to identify it

- " $f(x) = \dots$, find $f(3)$ ", "composite function $f(g(x))$ ", "inverse function", "domain".

How to solve it

- Evaluate by substituting the value of x .
- Composite $f(g(x))$: work from the inside out. Order matters: $f(g(x))$ is usually not $g(f(x))$.
- Inverse: write $y = f(x)$, solve for x in terms of y , then swap the letters.
- Domain: exclude values that make a denominator zero or give the square root of a negative number.

Example 1. If $f(x) = 2x + 3$ and $g(x) = x^2$, find $f(g(2))$ and $g(f(2))$.

Solution. $g(2) = 4$, so $f(g(2)) = f(4) = 11$.

$$f(2) = 7, \text{ so } g(f(2)) = g(7) = 49.$$

Example 2. Find the inverse of $f(x) = (3x - 2)/5$.

Solution. Put $y = (3x - 2)/5$. Then $5y = 3x - 2$, so $x = (5y + 2)/3$.

Swapping letters: $f^{-1}(x) = (5x + 2)/3$.

$$\text{Check: } f(1) = 1/5 \text{ and } f^{-1}(1/5) = (1 + 2)/3 = 1.$$

1.14 Permutation & Combination

1.14.1 Fundamental counting principle

What it is: If one task can be done in m ways and a second task in n ways, doing both one after the other can be done in $m \times n$ ways (AND means multiply). If either task alone will do, the count is $m + n$ (OR means add).

How to identify it

- "In how many ways can ... and ...", number of routes, outfits, passwords, numbers formed from digits.
- Digit questions with restrictions (no repetition, even number, first digit not zero).

How to solve it

- Break the job into stages and multiply the choices at each stage.
- Fill the most restricted place first (for an even number, fill the units digit first).
- When the cases split (such as units digit 0 versus non-zero), count each case separately and add.

Example 1. How many 3-digit numbers can be formed from the digits 1 to 5 if no digit is repeated?

Solution. Hundreds place: 5 choices. Tens place: 4 choices. Units place: 3 choices.

Total = $5 \times 4 \times 3 = 60$.

Example 2. How many 3-digit even numbers with all digits different can be formed from the digits 0 to 9?

Solution. Fill the units place first, in two cases.

Units digit 0: hundreds 9 choices, tens 8 choices, giving 72.

Units digit 2, 4, 6 or 8 (4 choices): hundreds cannot be 0 or the used digit, so 8 choices; tens has 8 choices left (0 is allowed). That gives $4 \times 8 \times 8 = 256$.

Total = $72 + 256 = 328$.

1.14.2 Permutations

What it is: Permutations count arrangements in which the order matters. Selecting r items from n and arranging them: $nPr = n!/(n - r)!$.

How to identify it

- "Arrange", "seated in a row", "rank", "form words", "positions", "president and secretary".

How to solve it

- All n different items in a row: $n!$.
- Letters with repeats: $n!/(p! \times q! \times \dots)$, where p, q are the counts of repeated letters.
- Items that must stay together: treat them as one block, arrange the blocks, then multiply by the internal arrangements.
- Items that must not be together: total arrangements – arrangements with them together.

Example 1. In how many ways can the letters of the word LEVEL be arranged?

Solution. LEVEL has 5 letters, with L twice and E twice.

Arrangements = $5!/(2! \times 2!) = 120/4 = 30$.

Example 2. In how many ways can 5 people sit in a row if two particular people must sit together?

Solution. Treat the pair as one block, so there are 4 items to arrange: $4! = 24$.

The pair can swap inside the block in $2! = 2$ ways.

Total = $24 \times 2 = 48$.

1.14.3 Combinations

What it is: Combinations count selections in which the order does not matter. $nCr = n!/[r!(n-r)!]$ and $nCr = nC(n-r)$.

How to identify it

- "Choose", "select", "committee", "team", "handshakes", "number of triangles or diagonals".

How to solve it

- Select from separate groups: multiply the combinations for each group.
- "At least one" = total selections – selections with none.
- Handshakes among n people: $nC2$. Diagonals of an n -sided polygon: $nC2 - n$.
- Choose first, arrange later: if the order matters afterwards, multiply the combinations by $r!$.

Example 1. A committee of 4 is to be formed from 6 men and 4 women, with exactly 2 women. In how many ways?

Solution. Choose 2 women from 4: $4C2 = 6$. Choose 2 men from 6: $6C2 = 15$.

Total = $6 \times 15 = 90$.

Example 2. A team of 3 is chosen from 5 men and 4 women and must include at least one woman. In how many ways?

Solution. Total teams = $9C3 = 84$.

Teams with no woman (all men) = $5C3 = 10$.

Teams with at least one woman = $84 - 10 = 74$.

1.14.4 Circular permutations

What it is: Arrangements around a circle, where rotating everyone one place gives the same arrangement. n different items can be arranged in $(n-1)!$ ways.

How to identify it

- "Seated around a round table", "arranged in a circle", "necklace", "garland".

How to solve it

- n different people around a table: $(n-1)!$. Fix one person to remove the rotations, then arrange the rest.
- Necklace or garland (it can be flipped over): $(n-1)!/2$.
- Alternate arrangement of two groups: fix and arrange the first group circularly, then place the second group in the gaps.

Example 1. In how many ways can 6 people sit around a round table?

Solution. Fix one person. The other 5 can be arranged in $5! = 120$ ways.

Example 2. In how many ways can 5 men and 5 women sit around a round table so that no two men are together?

Solution. Seat the men first: $(5-1)! = 24$ ways. This leaves 5 gaps between them.

Seat the 5 women in the gaps: $5! = 120$ ways.

Total = $24 \times 120 = 2,880$.

1.14.5 Distribution problems

What it is: Counting the ways to share objects among people or to split them into groups.

How to identify it

- "Distribute", "in how many ways can 10 sweets be given to 3 children", "divide into groups".
- Check whether the objects are identical or different, and whether the groups are labelled.

How to solve it

- Identical objects among r different people, anyone may get none: $(n+r-1)C(r-1)$.
- Identical objects, each person at least one: $(n-1)C(r-1)$.
- Different objects among r people (any number each): r^n .
- Different objects into groups of sizes a, b, c : $n!/(a! b! c!)$. If groups of equal size are unlabelled, also divide by the factorial of the number of equal groups.

Example 1. In how many ways can 10 identical sweets be given to 3 children so that each child gets at least one?

Solution. Use $(n-1)C(r-1) = 9C2$.

$9C2 = 36$.

Example 2. In how many ways can 6 different books be divided (a) among 3 people, 2 each, (b) into 3 unlabelled pairs?

Solution. (a) $6!/(2! 2! 2!) = 720/8 = 90$.

(b) The 3 pairs are not labelled, so divide by $3! = 6$: $90/6 = 15$.

1.15 Probability

1.15.1 Basic probability

What it is: Probability measures how likely an event is: $P(E) = (\text{number of favourable outcomes}) \div (\text{total number of equally likely outcomes})$. It lies between 0 and 1.

How to identify it

- "Probability", "chance", "at random", "likelihood".

How to solve it

- $P(E) = n(E)/n(S)$.
- $P(\text{not } E) = 1 - P(E)$. For "at least one", use $1 - P(\text{none})$.
- When selecting items, count favourable and total outcomes with combinations.

Example 1. A bag has 5 red and 3 blue balls. One ball is drawn at random. Find the probability that it is red.

Solution. Favourable = 5, total = 8.

$P(\text{red}) = 5/8$.

Example 2. Two balls are drawn together from the same bag (5 red, 3 blue). Find the probability that both are red.

Solution. Favourable = $5C2 = 10$. Total = $8C2 = 28$.

$P = 10/28 = 5/14$.

1.15.2 Independent & dependent events

What it is: Events are independent when one does not change the probability of the other (for example, with replacement). They are dependent when it does (for example, without replacement).

How to identify it

- "With replacement" or separate trials, such as two people each shooting: independent.
- "Without replacement" or "one after another from the same set": dependent.

How to solve it

- Independent: $P(A \text{ and } B) = P(A) \times P(B)$.
- Either event: $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$. For mutually exclusive events there is nothing to subtract.
- Dependent: multiply the first probability by the new (changed) probability of the second.
- "At least one occurs" = $1 - P(\text{none occurs})$.

Example 1. A hits a target with probability $1/3$ and B with probability $1/4$. Both fire once. Find the probability that the target is hit.

Solution. $P(\text{both miss}) = (1 - 1/3)(1 - 1/4) = 2/3 \times 3/4 = 1/2$.

$P(\text{target hit}) = 1 - 1/2 = 1/2$.

Example 2. Two cards are drawn one after another, without replacement, from a pack of 52. Find the probability that both are aces.

Solution. $P(\text{first ace}) = 4/52$. $P(\text{second ace after one removed}) = 3/51$.

$P = 4/52 \times 3/51 = 12/2652 = 1/221$.

1.15.3 Conditional probability

What it is: The probability of an event A when we already know that event B has happened: $P(A|B) = P(A \text{ and } B)/P(B)$.

How to identify it

- "Given that", "if it is known that", "provided that", "what is the probability that it came from..."

How to solve it

- Shrink the sample space to B, then count how much of B is also A.
- Formula: $P(A|B) = n(A \cap B)/n(B)$.
- Bayes' theorem (reverse a condition): $P(A|B) = P(B|A) \times P(A)/P(B)$, where $P(B)$ adds up all the ways B can happen.

Example 1. A die is rolled. Given that the result is even, find the probability that it is greater than 3.

Solution. Even results: 2, 4, 6 (3 outcomes). Of these, greater than 3: 4 and 6 (2 outcomes).
 $P = 2/3$.

Example 2. Machine A makes 60% of the items and 2% of its items are defective. Machine B makes 40% and 5% of its items are defective. An item picked at random is defective. Find the probability that it came from B.

Solution. $P(\text{defective}) = 0.6 \times 0.02 + 0.4 \times 0.05 = 0.012 + 0.020 = 0.032$.

$P(\text{from B and defective}) = 0.020$.

$P(B | \text{defective}) = 0.020/0.032 = 5/8$.

1.15.4 Cards, dice & coins

What it is: Standard sample spaces that appear in many questions. You need to know them by heart to count quickly.

How to identify it

- "A pack of cards", "two dice", "three coins are tossed", "sum of the numbers".

How to solve it

- Coins: n coins give 2^n outcomes. Exactly r heads: nCr ways.
- Two dice: 36 outcomes. Number of ways for sum 2 to 12: 1, 2, 3, 4, 5, 6, 5, 4, 3, 2, 1.
- Pack: 52 cards, 4 suits of 13, 26 red and 26 black, 12 face cards (J, Q, K), 4 aces.
- For "or" questions with overlap, subtract the common cards or outcomes.

Example 1. Two dice are thrown. Find the probability that the sum is 8.

Solution. Favourable pairs: (2,6), (3,5), (4,4), (5,3), (6,2), which is 5.
 $P = 5/36$.

Example 2. Three coins are tossed. Find the probability of exactly 2 heads.

Solution. Total outcomes = $2^3 = 8$. Exactly 2 heads: $3C2 = 3$ outcomes.
 $P = 3/8$.

Example 3. One card is drawn from a pack of 52. Find the probability that it is a king or a heart.

Solution. 4 kings + 13 hearts – 1 king of hearts (counted twice) = 16 cards.
 $P = 16/52 = 4/13$.

1.16 Geometry

1.16.1 Lines & angles

What it is: The basic rules about angles formed by lines, including parallel lines cut by a transversal.

How to identify it

- "Parallel lines", "transversal", "complementary", "supplementary", "find the angle".

How to solve it

- Complementary angles add to 90° . Supplementary angles add to 180° . A linear pair adds to 180° .
- Vertically opposite angles are equal. Angles around a point add to 360° .
- Parallel lines with a transversal: corresponding angles equal, alternate angles equal, co-interior (same-side) angles add to 180° .
- Always put the unknown as x, write the relation, and solve.

Example 1. An angle is 20° more than its complement. Find the angle.

Solution. Let the angle be x. Its complement is $90 - x$.
 $x = (90 - x) + 20$, so $2x = 110$ and $x = 55^\circ$.

Example 2. Two parallel lines are cut by a transversal. Two co-interior angles are $(3x + 10)^\circ$ and $(2x + 20)^\circ$. Find x and the angles.

Solution. Co-interior angles add to 180° : $5x + 30 = 180$, so $x = 30$.
The angles are 100° and 80° .

1.16.2 Triangles

What it is: A triangle has three sides and its angles add to 180° . Key topics are the Pythagoras theorem, area, similarity and special points (centroid, incentre, circumcentre).

How to identify it

- "Right-angled triangle", "area of a triangle with sides...", "similar triangles", "inradius", "median".

How to solve it

- Angle sum = 180° . An exterior angle = the sum of the two opposite interior angles.
- Pythagoras: $a^2 + b^2 = c^2$. Common triples: (3,4,5), (5,12,13), (8,15,17), (7,24,25).
- Area = $\frac{1}{2} \times \text{base} \times \text{height}$. Heron: $\sqrt{s(s-a)(s-b)(s-c)}$ with s = half the perimeter. Equilateral: $(\sqrt{3}/4)a^2$.
- Similar triangles: sides in proportion; ratio of areas = square of ratio of sides.
- Inradius = area $\div s$. Circumradius of a right triangle = half the hypotenuse. The centroid divides a median 2 : 1.

Example 1. Find the area of a triangle with sides 13 cm, 14 cm and 15 cm.

Solution. $s = (13 + 14 + 15)/2 = 21$.

Area = $\sqrt{21 \times 8 \times 7 \times 6} = \sqrt{7056} = 84 \text{ cm}^2$.

Example 2. Two similar triangles have sides in the ratio 2 : 3. The smaller has area 20 cm^2 . Find the area of the larger.

Solution. Ratio of areas = $2^2 : 3^2 = 4 : 9$.

Larger area = $20 \times 9/4 = 45 \text{ cm}^2$.

1.16.3 Circles

What it is: A circle is the set of points at a fixed distance (radius) from a centre. Questions use chords, tangents, arcs, sectors and the angles they make.

How to identify it

- "Chord", "tangent", "arc", "sector", "angle subtended", "distance from the centre".

How to solve it

- Circumference = $2\pi r$. Area = πr^2 .
- Arc length = $(\theta/360) \times 2\pi r$. Sector area = $(\theta/360) \times \pi r^2$.
- A perpendicular from the centre bisects a chord. Chord = $2\sqrt{r^2 - d^2}$ at distance d from the centre.
- A tangent is perpendicular to the radius at the point of contact. Tangents from one outside point are equal in length.
- The angle at the centre is twice the angle at the circumference on the same arc. The angle in a semicircle is 90° .

Example 1. A chord 16 cm long is in a circle of radius 10 cm. Find its distance from the centre.

Solution. The perpendicular from the centre bisects the chord, giving half-length 8 cm.

$d = \sqrt{10^2 - 8^2} = \sqrt{36} = 6 \text{ cm}$.

Example 2. Find the arc length and area of a sector with angle 72° in a circle of radius 10 cm (take $\pi = 3.14$).

Solution. Fraction of the circle = $72/360 = 1/5$.

Arc length = $1/5 \times 2 \times 3.14 \times 10 = 12.56 \text{ cm}$.

Area = $1/5 \times 3.14 \times 100 = 62.8 \text{ cm}^2$.

1.16.4 Quadrilaterals & polygons

What it is: Four-sided figures (square, rectangle, parallelogram, rhombus, trapezium) and many-sided polygons, with their angle and area rules.

How to identify it

- "Regular polygon", "interior or exterior angle", "diagonals", "area of a rhombus or trapezium".

How to solve it

- Sum of interior angles of an n -sided polygon = $(n - 2) \times 180^\circ$. Each exterior angle of a regular polygon = $360^\circ/n$.
- Number of diagonals = $n(n - 3)/2$.
- Rhombus: diagonals bisect each other at right angles, area = $1/2 \times d_1 \times d_2$. Parallelogram area = base $\times$ height. Trapezium area = $1/2 \times (\text{sum of parallel sides}) \times \text{height}$.
- Rectangle diagonal = $\sqrt{l^2 + b^2}$. Square diagonal = $a\sqrt{2}$.

Example 1. Each exterior angle of a regular polygon is 30° . Find the number of sides and the sum of the interior angles.

Solution. Number of sides = $360/30 = 12$.

Interior angle sum = $(12 - 2) \times 180 = 1800^\circ$.

Example 2. The diagonals of a rhombus are 16 cm and 12 cm. Find its side and area.

Solution. The diagonals bisect at right angles, giving half-diagonals 8 and 6.

Side = $\sqrt{8^2 + 6^2} = 10$ cm. Area = $1/2 \times 16 \times 12 = 96 \text{ cm}^2$.

1.16.5 Coordinate geometry

What it is: Describing points and lines with coordinates (x, y) on a plane and using formulas to find distances, slopes and areas.

How to identify it

- Points such as $A(2, 3)$, "distance between", "midpoint", "slope", "equation of a line", "area of a triangle with vertices..."

How to solve it

- Distance = $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$. Midpoint = $((x_1 + x_2)/2, (y_1 + y_2)/2)$.
- Section formula, dividing in ratio $m : n$ internally: $((m x_2 + n x_1)/(m + n), (m y_2 + n y_1)/(m + n))$.
- Slope $m = (y_2 - y_1)/(x_2 - x_1)$. Parallel lines have equal slopes; perpendicular lines have slopes whose product is -1 .
- Line: $y = mx + c$, or $y - y_1 = m(x - x_1)$.
- Area of a triangle = $1/2 \times |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$.

Example 1. Find the distance between $(1, 2)$ and $(7, 10)$, and the midpoint.

Solution. Distance = $\sqrt{6^2 + 8^2} = \sqrt{100} = 10$.

Midpoint = $((1 + 7)/2, (2 + 10)/2) = (4, 6)$.

Example 2. Find the area of the triangle with vertices $(1, 1)$, $(5, 1)$ and $(3, 5)$.

Solution. Area = $1/2 \times |1(1 - 5) + 5(5 - 1) + 3(1 - 1)| = 1/2 \times |-4 + 20 + 0| = 8$.

Check: base 4 and height 4 give $1/2 \times 4 \times 4 = 8$.

1.17 Mensuration

1.17.1 2D figures: area & perimeter

What it is: Area is the surface a flat figure covers; perimeter is the length of its boundary.

How to identify it

- "Area", "perimeter", "cost of fencing or carpeting", "a path around a garden", "wire bent into a shape".

How to solve it

- Rectangle: area = $l \times b$, perimeter = $2(l + b)$. Square: a^2 and $4a$. Triangle: $\frac{1}{2} \times b \times h$. Circle: πr^2 and $2\pi r$.
- Parallelogram: $b \times h$. Trapezium: $\frac{1}{2} \times (a + b) \times h$. Rhombus: $\frac{1}{2} \times d_1 \times d_2$.
- A path of uniform width: area of path = outer area – inner area.
- Cost = area $\times$ rate per unit area (carpet) or perimeter $\times$ rate per unit length (fencing).
- If a wire is reshaped, its length (perimeter) stays the same.

Example 1. A rectangular park 60 m by 40 m has a 2 m wide path running inside along its border. Find the area of the path.

Solution. Inner rectangle = $(60 - 4) \times (40 - 4) = 56 \times 36 = 2016 \text{ m}^2$.

Outer = $60 \times 40 = 2400 \text{ m}^2$.

Path = $2400 - 2016 = 384 \text{ m}^2$.

Example 2. A circular wire of radius 14 cm is bent into a square. Find the side and the area of the square ($\pi = 22/7$).

Solution. Length of wire = $2 \times 22/7 \times 14 = 88 \text{ cm}$.

Side = $88 \div 4 = 22 \text{ cm}$. Area = $22^2 = 484 \text{ cm}^2$.

1.17.2 3D figures: surface area & volume

What it is: Surface area is the total area of the faces of a solid; volume is the space it occupies.

How to identify it

- "Volume", "capacity", "curved surface area", "melted and recast", "cube, cuboid, cylinder, cone, sphere".

How to solve it

- Cube: $V = a^3$, TSA = $6a^2$, diagonal = $a\sqrt{3}$. Cuboid: $V = lbh$, TSA = $2(lb + bh + hl)$, diagonal = $\sqrt{l^2 + b^2 + h^2}$.
- Cylinder: $V = \pi r^2 h$, CSA = $2\pi rh$, TSA = $2\pi r(r + h)$.
- Cone: $V = \frac{1}{3} \pi r^2 h$, CSA = πrl , slant height $l = \sqrt{r^2 + h^2}$.
- Sphere: $V = \frac{4}{3} \pi r^3$, surface area = $4\pi r^2$. Hemisphere: $V = \frac{2}{3} \pi r^3$, TSA = $3\pi r^2$.
- Melting and recasting: the volume stays the same, so equate the volumes.

Example 1. A cylinder has radius 7 cm and height 10 cm. Find its volume and total surface area ($\pi = 22/7$).

Solution. Volume = $22/7 \times 49 \times 10 = 1540 \text{ cm}^3$.

TSA = $2 \times 22/7 \times 7 \times (7 + 10) = 44 \times 17 = 748 \text{ cm}^2$.

Example 2. A metal sphere of radius 6 cm is melted and recast into small spheres of radius 2 cm. How many small spheres are made?

Solution. Volumes are in the ratio of the cubes of the radii.

Number = $(6/2)^3 = 27$.

1.18 Trigonometry

1.18.1 Ratios & identities

What it is: Trigonometric ratios relate the sides of a right-angled triangle to its angles: $\sin = \text{opposite/hypotenuse}$, $\cos = \text{adjacent/hypotenuse}$, $\tan = \text{opposite/adjacent}$.

How to identify it

- "If $\sin \theta = 3/5$, find...", "value of $\sin 30^\circ + \cos 60^\circ$ ", "prove that...", "simplify".

How to solve it

- Identities: $\sin^2 \theta + \cos^2 \theta = 1$, $1 + \tan^2 \theta = \sec^2 \theta$, $1 + \cot^2 \theta = \text{cosec}^2 \theta$. Also $\tan \theta = \sin \theta / \cos \theta$.
- Standard values at 0° , 30° , 45° , 60° , 90° : $\sin = 0, 1/2, 1/\sqrt{2}, \sqrt{3}/2, 1$. $\cos$ is the same list reversed. $\tan = 0, 1/\sqrt{3}, 1, \sqrt{3}, \text{undefined}$.
- Complementary angles: $\sin(90^\circ - \theta) = \cos \theta$ and $\tan(90^\circ - \theta) = \cot \theta$.
- Given one ratio, draw the right triangle, find the third side by Pythagoras, then read off the others.

Example 1. If $\sin \theta = 3/5$ and θ is acute, find $\cos \theta$ and $\tan \theta$.

Solution. Opposite = 3, hypotenuse = 5, so adjacent = $\sqrt{25 - 9} = 4$.
 $\cos \theta = 4/5$ and $\tan \theta = 3/4$.

Example 2. Find the value of $\sin^2 30^\circ + \cos^2 60^\circ + \tan 45^\circ$.

Solution. $\sin 30^\circ = 1/2$, so $\sin^2 30^\circ = 1/4$. $\cos 60^\circ = 1/2$, so $\cos^2 60^\circ = 1/4$. $\tan 45^\circ = 1$.
Total = $1/4 + 1/4 + 1 = 1.5$.

1.18.2 Heights & distances

What it is: Using angles of elevation and depression with right triangles to find heights of towers, lengths of shadows and distances.

How to identify it

- "Angle of elevation", "angle of depression", "height of a tower or pole", "length of shadow", "from a point at a distance of..."

How to solve it

- Draw a clear right triangle. Angles of elevation and depression are always measured from the horizontal.
- The angle of depression from the top equals the angle of elevation from the bottom (alternate angles).
- $\tan \theta = \text{height} \div \text{horizontal distance}$. Use $\sin$ or $\cos$ when the slanting length is involved.
- With two observation points, write two equations with the same height and solve together.

Example 1. From a point 30 m from the foot of a tower, the angle of elevation of its top is 60° . Find the height.

Solution. $\tan 60^\circ = h/30$.

$h = 30 \times \sqrt{3} = 30\sqrt{3} \approx 51.96$ m.

Example 2. From the top of a 50 m cliff, the angle of depression of a boat is 45° . How far is the boat from the foot of the cliff?

Solution. $\tan 45^\circ = 50/d$, and $\tan 45^\circ = 1$.

$d = 50$ m.

Example 3. The angle of elevation of the top of a tower from a point on the ground is 30° . After walking 20 m towards the tower it becomes 45° . Find the height.

Solution. Let the first distance be d . Then $h = d \tan 30^\circ = d/\sqrt{3}$, and $h = (d - 20) \tan 45^\circ = d - 20$.

$d/\sqrt{3} = d - 20$, so $d(1 - 1/\sqrt{3}) = 20$ and $d = 20\sqrt{3}/(\sqrt{3} - 1) = 10\sqrt{3}(\sqrt{3} + 1) = 30 + 10\sqrt{3}$.

$h = d - 20 = 10 + 10\sqrt{3} \approx 27.3$ m.

1.19 Calendar & Clock

1.19.1 Calendar

What it is: Finding the day of the week for any date using the idea of odd days (the remainder when days are divided by 7).

How to identify it

- "What day of the week was 15 August 1947", "which year has the same calendar", "how many Sundays in a month", "leap year".

How to solve it

- Odd days = (total days) mod 7. An ordinary year has 1 odd day, a leap year has 2.
- A year is a leap year if divisible by 4, but a century year must be divisible by 400 (1900 is not, 2000 is).
- Odd days in 100 years = 5, in 200 years = 3, in 300 years = 1, in 400 years = 0.
- Months with 31 days have 3 odd days, 30 days have 2, February has 0 (28 days) or 1 (29 days).
- Count odd days up to the day before the date. 0 = Sunday, 1 = Monday, ... 6 = Saturday.

Example 1. Which day of the week was 15 August 1947?

Solution. Complete years up to 1946: 1600 years give 0, the next 300 years give 1.

The remaining 46 years have 11 leap years, so $46 + 11 = 57$ odd days, which is 1 odd day. Total so far = $0 + 1 + 1 = 2$ (a Tuesday, 31 December 1946).

Days in 1947 up to 15 August = $31 + 28 + 31 + 30 + 31 + 30 + 31 + 15 = 227$, and $227 \bmod 7 = 3$.

Odd days = $2 + 3 = 5$, so the day was a Friday.

Example 2. 1 January 2026 is a Thursday. Which day is 1 January 2027, and which of 1900, 2000, 2024 are leap years?

Solution. 2026 is an ordinary year with 1 odd day, so 1 January 2027 is a Friday.

1900 is not a leap year (a century not divisible by 400). 2000 is a leap year. 2024 is a leap year.

1.19.2 Clock angles

What it is: Finding the angle between the hour and minute hands, or the time at which they make a given angle.

How to identify it

- "Angle between the hands at 4:20", "at what time do the hands coincide, point opposite, or make a right angle".

How to solve it

- The minute hand moves 6° per minute. The hour hand moves 0.5° per minute (30° per hour).
- Angle at h hours and m minutes = $|30h - 5.5m|$. If the result is above 180° , subtract it from 360° .
- To find a time for a given angle, set $30h - 5.5m$ equal to the angle (try both $+\text{angle}$ and $-\text{angle}$) and solve for m .
- In 12 hours the hands coincide 11 times, are at right angles 22 times, and opposite 11 times.

Example 1. Find the angle between the hands at 4:20.

Solution. Angle = $|30 \times 4 - 5.5 \times 20| = |120 - 110| = 10^\circ$.

Example 2. At what time between 4 and 5 o'clock do the hands coincide?

Solution. Set $30 \times 4 - 5.5m = 0$, so $m = 120/5.5 = 240/11 = 21 \frac{9}{11}$.

The hands coincide at $21 \frac{9}{11}$ minutes past 4.

1.19.3 Fast & slow clocks

What it is: A faulty clock gains or loses time at a steady rate, so its reading differs from the true time by an amount that grows over time.

How to identify it

- "Gains 5 minutes per hour", "loses", "was set right at 6 a.m.; what is the true time when it shows...", "hands coincide every 64 minutes".

How to solve it

- A clock gaining g minutes each true hour shows $(60 + g)$ minutes for every 60 true minutes. So true time = shown time $\times 60/(60 + g)$.
- Hands of a correct clock coincide every $65 \frac{5}{11}$ minutes ($720/11$).
- If they coincide every x minutes of true time: when x is less than $720/11$ the clock gains, when it is more the clock loses. The gain or loss per coincidence = $|720/11 - x|$.

- Multiply by the number of coincidences to find the gain or loss over a long period.

Example 1. A watch gains 4 minutes every hour. It is set right at 6 a.m. What is the true time when the watch shows 6 p.m.?

Solution. The watch shows 12 hours = 720 minutes.

It shows 64 minutes for every 60 true minutes, so true elapsed time = $720 \times 60/64 = 675$ minutes = 11 h 15 min.

True time = 6 a.m. + 11 h 15 min = 5:15 p.m.

Example 2. The hands of a clock coincide every 64 minutes of true time. Does the clock gain or lose, and how much in 24 hours?

Solution. On a correct clock they coincide every $720/11 = 65 \frac{5}{11}$ minutes, which is more than 64, so the clock gains.

Gain per coincidence = $720/11 - 64 = 16/11$ minutes.

Coincidences in 24 hours = $1440/64 = 22.5$.

Gain in a day = $22.5 \times 16/11 = 360/11 = 32 \frac{8}{11}$ minutes.

1.20 Set Theory & Venn Diagrams

1.20.1 Union, intersection & complement

What it is: A set is a collection of items. The union $A \cup B$ has the items in A or B (or both), the intersection $A \cap B$ has the items in both, and the complement A' has everything in the universal set that is not in A.

How to identify it

- "Play cricket or football", "play both", "neither", "only cricket".

How to solve it

- $n(A \cup B) = n(A) + n(B) - n(A \cap B)$.
- Neither = total - $n(A \cup B)$. Complement: $n(A') = n(U) - n(A)$.
- Only A = $n(A) - n(A \cap B)$.
- Draw two overlapping circles and fill the overlap first.

Example 1. In a class of 50, 30 play cricket, 25 play football and 10 play both. How many play neither?

Solution. $n(A \cup B) = 30 + 25 - 10 = 45$.

Neither = $50 - 45 = 5$.

Example 2. Of 60 students, 35 take Maths, 28 take Physics and 8 take neither. How many take both?

Solution. Students taking at least one subject = $60 - 8 = 52$.

$52 = 35 + 28 - \text{both}$, so both = $63 - 52 = 11$.

1.20.2 Overlapping sets

What it is: Counting with three overlapping groups, where regions of the Venn diagram are shared by two or three sets.

How to identify it

- "Read A, B or C", "exactly two", "exactly one", "at least one", three-circle problems.

How to solve it

- $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$.
- Start from the centre (all three) and work outward: each pair-only region = pair total - centre.
- Exactly two = (sum of pair intersections) - $3 \times$ (all three).
- Exactly one = at least one - exactly two - all three.

Example 1. In a survey of 100 people, 50 read paper A, 45 read B and 40 read C. 20 read A and B, 15 read B and C, 10 read A and C, and 5 read all three. How many read at least one paper, and how many read none?

Solution. At least one = $50 + 45 + 40 - 20 - 15 - 10 + 5 = 95$.

None = $100 - 95 = 5$.

Example 2. For the same survey, how many read exactly two papers and how many read exactly one?

Solution. Exactly two = $(20 + 15 + 10) - 3 \times 5 = 30$.

Exactly one = $95 - 30 - 5 = 60$.

Check by regions: only A = 25, only B = 15, only C = 20, total 60.

1.21 Other Quantitative Topics

1.21.1 Number series

What it is: A list of numbers that follow a rule. You must find the next term or the wrong term.

How to identify it

- "Find the next number", "what comes in place of ?", "find the wrong number".

How to solve it

- Take the differences between terms. If these are not constant, take differences again (second order).
- Check ratios (multiply by a constant), then squares, cubes, primes and patterns like $\times 2 + 1$.
- Two interleaved series: separate the odd and even positions.
- For a wrong term, find the pattern from the other terms and see which one breaks it.

Example 1. Find the next term: 3, 8, 15, 24, 35, ?

Solution. Differences: 5, 7, 9, 11, so the next difference is 13.

Next term = $35 + 13 = 48$. (The pattern is $n^2 - 1$: $2^2 - 1, 3^2 - 1, \dots, 7^2 - 1 = 48$.)

Example 2. Find the next term: 2, 3, 5, 9, 17, ?

Solution. Differences: 1, 2, 4, 8, so the next difference is 16.

Next term = $17 + 16 = 33$.

Example 3. Find the wrong number: 4, 9, 16, 26, 36, 49.

Solution. The numbers are squares: $2^2, 3^2, 4^2, 5^2, 6^2, 7^2$.

26 should be 25, so 26 is the wrong number.

1.21.2 Stocks, shares & banker's discount

What it is: Shares have a face value (printed) and a market value (current price); dividend is paid on face value.

Banker's discount is the interest charged by a bank on the face value of a bill paid early; true discount is the interest on its present worth.

How to identify it

- "Shares", "dividend", "brokerage", "at a premium or discount", "bill due after...", "banker's discount", "true discount".

How to solve it

- Number of shares = money invested $\div$ market price per share.
- Dividend per share = dividend % $\times$ face value. Income = number of shares $\times$ dividend per share.
- Return % on investment = (dividend per share $\div$ market price) $\times$ 100.
- True discount $TD = F \times R \times T / (100 + R \times T)$ (F = face value of bill). Banker's discount $BD = F \times R \times T / 100$.
- $BD - TD$ = interest on $TD = TD^2 / PW$ (PW = present worth = $F - TD$).

Example 1. How much income does a person get by investing ₹12,000 in 10% shares of face value ₹100 at a market price of ₹120? What is the return percent?

Solution. Number of shares = $12000 \div 120 = 100$.

Income = $100 \times 10 = ₹1,000$.

Return = $1000 \div 12000 \times 100 = 8.33\%$.

Example 2. A bill of ₹1,100 is due in 1 year and the rate is 10% per annum. Find the true discount, the banker's discount and their difference.

Solution. $TD = 1100 \times 10 \div (100 + 10) = ₹100$. Present worth = $1100 - 100 = ₹1,000$.

$BD = 1100 \times 10 \div 100 = ₹110$.

Difference = ₹10, which agrees with $TD^2 / PW = 10000 / 1000 = 10$.

1.21.3 Data sufficiency

What it is: A question followed by statements. You decide whether the statements give enough information to answer, without necessarily computing the final answer.

How to identify it

- "Is statement I alone sufficient", "the question is followed by two statements I and II", options such as "both I and II together are needed".

How to solve it

- Check statement I alone, then II alone, then both together.
- For a value question, a statement is sufficient only if it gives exactly one value.
- For a yes/no question, a statement is sufficient if the answer is always yes or always no. If it is sometimes yes and sometimes no, it is not sufficient.
- Count unknowns versus independent equations as a quick check, and test with a few numbers.

Example 1. What is Ravi's age? Statement I: Ravi is 5 years older than Mohan. Statement II: Mohan is 20 years old.

Solution. I alone: Mohan's age is unknown, so Ravi's age cannot be found.

II alone: tells nothing about Ravi.

Together: $\text{Ravi} = 20 + 5 = 25$. Both statements are needed.

Example 2. Is the integer n even? Statement I: n^2 is divisible by 4. Statement II: n is a multiple of 3.

Solution. I: if n were odd, n^2 would be odd and could not be divisible by 4. So n is always even, which answers the question. Sufficient.

II: $n = 3$ (odd) and $n = 6$ (even) both fit, so no single answer. Not sufficient.

Answer: statement I alone is sufficient.

Part 2: Logical Reasoning

Tests how clearly you can think, not what you have memorised. Draw diagrams, write symbols, and work one clue at a time. Figure-based questions are described in words here.

2.1 Verbal Reasoning

2.1.1 Number & letter series

What it is: A sequence of numbers or letters that follows a hidden rule. You find the next term or the missing term.

How to identify it

- "Find the next term", "what comes in place of the question mark", sequences like A, C, F, J, ?
- Letter pairs such as AZ, BY, CX, ? (two series running together).

How to solve it

- Convert letters to positions (A = 1, B = 2, ... Z = 26). The opposite letter of a letter at position p is at $27 - p$ (A and Z, B and Y).
- Then treat the positions as a number series: look at the gaps, the second gaps, multiples, squares, or an alternating pattern.
- If two things change at once (such as first and second letters of a pair), solve each separately.

Example 1. Find the next letter: A, C, F, J, ?

Solution. Positions: 1, 3, 6, 10.

Gaps: 2, 3, 4, so the next gap is 5 and the next position is 15.

The letter is O.

Example 2. Find the next pair: AZ, BY, CX, ?

Solution. The first letters go A, B, C, so the next is D.

The second letters go Z, Y, X, so the next is W.

The next pair is DW.

2.1.2 Coding-decoding

What it is: Words or numbers are changed into a code by a fixed rule. You find the rule and apply it to a new word.

How to identify it

- "In a certain code, ... is written as How is ... written in that code?"
- Several coded sentences given, asking for the code of one word.

How to solve it

- Letter coding: check for a shift (+1, -2), reversal of the alphabet, reversal of the word, or shifts that grow with each letter. Test the rule on every letter.
- Word coding (sentences): compare two sentences that share a word. The common code number or letter belongs to the common word.
- Number coding: letters replaced by their positions, then added or multiplied.

Example 1. If COLD is written as DPME, how is WARM written?

Solution. C→D, O→P, L→M, D→E: every letter moves one step forward.

W→X, A→B, R→S, M→N.

WARM is written XBSN.

Example 2. In a code, "sea is deep" is 4 5 6, "deep sea water" is 6 4 8, and "is water cold" is 5 8 9. What is the code for "water"?

Solution. Sentences 1 and 2 share "sea" and "deep", so 4 and 6 are the codes of those two words.

Sentences 1 and 3 share only "is", so "is" = 5.

Sentences 2 and 3 share "water". The common code in 6 4 8 and 5 8 9 is 8, so "water" = 8.

2.1.3 Blood relations

What it is: Working out how two people in a family are related from a chain of clues.

How to identify it

- "A is the brother of B", "pointing to a photograph, she said...", "how is X related to Y?"
- Coded relations such as "A + B means A is the mother of B".

How to solve it

- Draw a family tree: squares for males, circles for females, a horizontal line for a married couple, a vertical line down to children.
- Move one clue at a time and note the gender whenever it is known.
- Translations: father's or mother's brother = uncle; father's or mother's sister = aunt; brother's son = nephew; sister's husband = brother-in-law; "my father's only son" = myself or my brother depending on context.
- For coded relations, replace each symbol by its meaning first, then draw the tree.

Example 1. Pointing to a man, Rina said, "He is the son of my grandfather's only son." How is the man related to Rina?

Solution. Rina's grandfather's only son is Rina's father.
That man's son is Rina's brother.
He is Rina's brother.

Example 2. A is B's sister. C is B's mother. D is C's father. E is D's mother. How is A related to D, and to E?

Solution. B's mother is C, and A is B's sister, so A is C's daughter.
D is C's father, so D is A's grandfather, and A is D's granddaughter.
E is D's mother, so A is E's great-granddaughter.

2.1.4 Direction sense

What it is: Tracking movement and turns to find the final direction or the distance from the starting point.

How to identify it

- "Walks 5 km north, then turns right...", "which direction is he facing now", "how far is he from the starting point".
- Shadow problems at sunrise or sunset.

How to solve it

- Draw a grid with north at the top. A right turn is 90° clockwise, a left turn is 90° anticlockwise from the direction you are facing.
- Mark each leg and its length. The final straight-line distance uses Pythagoras when the legs are at right angles.
- Shadows: in the morning the sun is in the east, so shadows fall to the west. In the evening shadows fall to the east.

Example 1. A man walks 6 km north, turns right and walks 8 km, then turns right again and walks 6 km. How far and in which direction is he from the starting point?

Solution. North 6 km. Right turn from north faces east: 8 km east. Right turn from east faces south: 6 km south. The north and south legs cancel, leaving 8 km east of the start.

Example 2. Anil walks 12 m south, turns left and walks 5 m. How far is he from the start and in which direction?

Solution. Facing south, a left turn faces east. So he is 12 m south and 5 m east of the start.
Distance = $\sqrt{(12^2 + 5^2)} = \sqrt{169} = 13$ m, in the south-east direction.

2.1.5 Ranking & order

What it is: Finding a person's position or the total count from their rank counted from different ends of a line.

How to identify it

- "10th from the top and 15th from the bottom", "how many students are there", "how many are between".
- Positions that change when two people exchange places.

How to solve it

- Total = rank from one end + rank from the other end – 1.
- Rank from the other end = total – rank + 1.
- People between positions p and q = $|p - q| - 1$.
- Exchange problems: the first person moves into the second person's old place. Use the new and old positions to link the two ends.

Example 1. Rahul is 12th from the top and 20th from the bottom in a class. How many students are there?

Solution. Total = $12 + 20 - 1 = 31$.

Example 2. In a row, A is 9th from the left and B is 11th from the right. After they exchange places, A is 20th from the left. How many people are in the row?

Solution. After the exchange A stands in B's old place, which is 20th from the left.

B's old place is 11th from the right, so total = $20 + 11 - 1 = 30$.

2.1.6 Seating arrangement

What it is: Placing a group of people in a row, around a table or in a block according to given conditions.

How to identify it

- "Sit in a row facing north", "sit around a circular table facing the centre", "second to the left of", "opposite to".

How to solve it

- Draw the layout and number the seats.
- Start with the most definite clues (a fixed seat, an end seat, "opposite"), then place the rest.
- Row facing north: left is west (towards the left of the page), right is east. Circle facing the centre: a person's left is the next seat clockwise, and right is the next anticlockwise. If they face outwards, reverse this.
- When a clue leaves two cases open, keep both and use later clues to eliminate one.

Example 1. Five friends A, B, C, D, E sit in a row facing north. A is at the extreme left. B sits immediately to the right of A. C does not sit next to B. D sits immediately to the left of E. Who sits in the middle?

Solution. Seats 1 to 5 from the left. A is in seat 1 and B is in seat 2.

C is not next to B, so C is in seat 4 or 5. D and E take consecutive seats with D on the left: seats (3, 4) or (4, 5).

If D and E take seats 4 and 5, C would have no seat that is not next to B, so D and E are in seats 3 and 4 and C is in seat 5.

Order: A B D E C. D sits in the middle.

Example 2. Six people A, B, C, D, E, F sit around a circular table facing the centre. A is opposite D. B is on the immediate right of A. C is second to the left of A. E is not next to C. Who sits between A and C?

Solution. Number the seats 1 to 6 clockwise and put A in seat 1. D is opposite, in seat 4.

Right of A is anticlockwise, so B is in seat 6. Second to the left is clockwise, so C is in seat 3.

Seats 2 and 5 are left for E and F. Seat 2 is next to C (seat 3), so E takes seat 5 and F takes seat 2.

F sits between A and C.

2.1.7 Puzzles

What it is: Logic problems with several people or things and many conditions (floors, days, pets, colours, heights).

How to identify it

- "Four friends each own a different pet", "seven people live on seven floors", "who is the tallest".

How to solve it

- Make a table with the people as rows and the attribute (pet, colour, day) as columns.
- Enter the definite clues first. Cross out every option a negative clue removes.
- Look for a row or column with one option left and fill it in; repeat.
- For comparison clues (taller, older, heavier), build a chain with $>$ and $<$ symbols and join clues that share a name.

Example 1. Four friends A, B, C, D own a cat, dog, fish and bird (one each, all different). A owns neither the cat nor the dog. B owns the cat. C does not own the fish. D owns the dog. Who owns what?

Solution. B owns the cat and D owns the dog.

A cannot have cat or dog, so A has the fish or the bird. C cannot have the fish, so C has the bird.

That leaves the fish for A.

A: fish, B: cat, C: bird, D: dog.

Example 2. P is taller than Q. R is shorter than Q but taller than S. T is taller than P. Who is the tallest and who is the shortest?

Solution. $T > P$ and $P > Q$ and $Q > R$ and $R > S$.

Chain: $T > P > Q > R > S$.

T is the tallest and S is the shortest.

2.1.8 Syllogisms

What it is: Drawing conclusions from statements about groups, using the logic of sets (All, No, Some, Some not).

How to identify it

- "Statements: All A are B. Some B are C. Conclusions: I... II..."
- Statements with "all", "no", "some", "some... are not".

How to solve it

- Draw a Venn diagram for each statement. A conclusion follows only if it is true in every possible diagram.
- Useful rules: All A are B + All B are C gives All A are C. Some A are B + All B are C gives Some A are C. No A is B + All C are A gives No C is B. All A are B + Some B are C gives no sure conclusion. Some + Some gives nothing.
- Conversion: "No A is B" gives "No B is A". "Some A are B" gives "Some B are A". "All A are B" gives only "Some B are A".
- If two conclusions are a "Some" and a "No" (or "All" and "Some not") about the same terms, then "either I or II" follows.

Example 1. Statements: All pens are books. All books are tables. Conclusions: I. All pens are tables. II. Some tables are books.

Solution. Pens are inside books and books are inside tables, so all pens are tables. Conclusion I follows.
"All books are tables" converts to "Some tables are books". Conclusion II follows.
Both follow.

Example 2. Statements: Some cats are dogs. All dogs are birds. Conclusions: I. Some cats are birds. II. All cats are birds.

Solution. The cats that are dogs are inside the dogs, which are inside the birds, so some cats are birds. Conclusion I follows.
Nothing says every cat is a dog, so II does not follow.
Only I follows.

2.1.9 Statement & conclusion

What it is: Deciding which conclusions can be drawn directly from a given statement, using only the information in it.

How to identify it

- "Statement: ... Conclusions: I... II..." with everyday sentences, not letters.

How to solve it

- Ignore everything you know from outside the statement.
- A conclusion follows if it is stated or must be true. It does not follow if it adds a new fact, predicts the future, or makes a stronger claim than the statement (for example, "all" instead of "some").
- Check each conclusion separately.
- If two conclusions cover opposite possibilities and neither can be decided, "either I or II" may follow.

Example 1. Statement: Only candidates who scored above 60 percent were called for the interview. Rahul was called for the interview. Conclusions: I. Rahul scored above 60 percent. II. Rahul will be selected.

Solution. Being called required scoring above 60 percent, so I follows.
The statement says nothing about the result of the interview, so II does not follow.
Only I follows.

Example 2. Statement: In the last survey, 70% of the voters in the town supported the new plan. Conclusions: I. Some voters did not support the plan. II. The town council will accept the plan.

Solution. 70% supported it, so up to 30% did not. Conclusion I follows.
The statement is silent about what the council will do, so II does not follow.
Only I follows.

2.1.10 Statement & assumption

What it is: An assumption is something the speaker takes for granted. The statement would make no sense if the assumption were false.

How to identify it

- "Statement: ... Assumptions: I... II...", "which assumption is implicit in the statement".

How to solve it

- Ask: "Is the speaker relying on this being true?" If yes, it is implicit.
- Negation test: if the opposite of the assumption makes the statement pointless, the assumption is implicit.
- An assumption is not a conclusion, a repeat of the statement, or an extra detail.
- Be careful of extreme words (only, always, never, all, definitely). These are usually not implicit.

Example 1. Statement: "Please carry an umbrella with you; it may rain today." Assumptions: I. An umbrella protects a person from rain. II. It will certainly rain today.

Solution. I is implicit: the advice only makes sense if umbrellas keep you dry.

II contradicts the word "may" and is too strong.

Only I is implicit.

Example 2. Statement: "The company advertised its new product on TV to increase sales." Assumptions: I. People watch TV advertisements. II. Sales never increase without advertising.

Solution. I is implicit: advertising on TV is pointless if nobody watches it.

II is an extreme claim that the statement does not need.

Only I is implicit.

2.1.11 Statement & argument

What it is: Judging whether an argument for or against a proposal is strong or weak.

How to identify it

- "Statement: Should... ? Arguments: I. Yes, because... II. No, because..."

How to solve it

- A strong argument is directly related to the question, important, and based on facts or sound reasoning.
- A weak argument is vague, based on opinion or emotion, about a small or personal matter, or says "because others do it".
- Judge each argument on its own. Both can be strong, both can be weak, or only one.
- Do not let your own opinion decide the answer; decide only on the quality of the argument.

Example 1. Statement: Should the government ban plastic bags? Arguments: I. Yes, they cause serious damage to the environment and block drains. II. No, some shopkeepers will lose a little revenue.

Solution. I is a direct and important reason, so it is strong.

II is a small private concern compared with the main issue, so it is weak.

Only I is strong.

Example 2. Statement: Should voting be made compulsory? Arguments: I. No, because some people do not like voting. II. Yes, because it ensures wider participation and a more representative government.

Solution. I states a personal preference, not a principle, so it is weak.

II is a relevant and important reason, so it is strong.

Only II is strong.

2.1.12 Course of action

What it is: Choosing which step or steps would properly deal with a problem described in a statement.

How to identify it

- "Statement: (a problem). Courses of action: I... II...", "which course of action should be followed".

How to solve it

- A suitable course of action is practical, relevant to the cause of the problem, and does no extra harm.
- Reject steps that are extreme (shut down everything), impractical, irrelevant, or that only blame someone.
- Both courses can follow if they are different and work together.
- If the problem is already solved or the action is a temporary fix of no value, it does not follow.

Example 1. Statement: Many accidents occur at a busy crossing because drivers ignore the traffic signals. Courses of action: I. Cameras should be installed to fine drivers who break the signal. II. The road should be closed permanently.

Solution. I is practical and discourages the cause of the accidents, so it follows.
II is an extreme step that would cause large inconvenience, so it does not follow.
Only I follows.

Example 2. Statement: The river water near the city has become polluted by waste from factories. Courses of action: I. The government should make factories treat their waste before releasing it. II. The government should shut down all factories.

Solution. I deals with the cause in a practical way, so it follows.
II is extreme and would harm jobs and the economy, so it does not follow.
Only I follows.

2.1.13 Cause & effect

What it is: Two statements describe events. You decide whether one is the cause of the other, both are effects of a common cause, or they are unrelated.

How to identify it

- "Two statements I and II are given. Which is the cause and which is the effect?"

How to solve it

- Ask which event would naturally happen first and bring about the other. A cause happens first (or exists independently).
- If both events look like results of some third, unstated event, they are effects of a common cause.
- If there is no connection, they are independent.
- Judge only from the statements, with common sense.

Example 1. I. The school declared a holiday. II. Heavy rain flooded the roads.

Solution. Flooded roads make a holiday necessary, not the other way round.
II is the cause and I is the effect.

Example 2. I. Shops in the city market stayed closed all day. II. Traffic on the city's roads was very light.

Solution. Neither causes the other. Both would result from something else, such as a city-wide strike.
Both are effects of a common cause.

2.1.14 Analogy

What it is: Finding the relationship between a given pair and applying the same relationship to another pair.

How to identify it

- "A : B :: C : ?", "Doctor : Hospital :: Teacher : ?", "8 : 512 :: 6 : ?"

How to solve it

- State the relationship between the first pair in one short sentence ("a person and the place they work").
- Apply that sentence to the second pair and check each option.
- Common types: worker and workplace, tool and user, part and whole, synonym or antonym, cause and effect, number rules (square, cube), letter shifts.
- For numbers, test the operations (square, cube, $\times n$, $+n$) and see which fits both.

Example 1. Doctor : Hospital :: Teacher : ? (School, Book, Student, Class)

Solution. A doctor works in a hospital, so a teacher works in a school.

Answer: School.

Example 2. 8 : 512 :: 6 : ?

Solution. $8^3 = 512$, so the rule is "cube the number".

$6^3 = 216$.

2.1.15 Classification (odd one out)

What it is: Finding the item that does not belong with the others because it breaks the common rule.

How to identify it

- "Choose the odd one out", groups of words, numbers or letter groups.

How to solve it

- Look for a property shared by all but one: category (fruit, flower), number property (prime, square, multiple of 3), pattern of letter gaps.
- For numbers, test prime/composite, odd/even, squares and cubes, divisibility, digit sum.
- For letter groups, compare the gap inside each group and the gap between groups.

Example 1. Find the odd one out: 3, 5, 7, 9, 11.

Solution. 3, 5, 7 and 11 are prime numbers.

$9 = 3 \times 3$ is not prime, so 9 is the odd one out.

Example 2. Find the odd one out: AC, EG, IK, MO, QT.

Solution. In AC, EG, IK and MO the second letter is 2 places after the first. The first letters are 4 places apart.

QT has a gap of 3 ($Q = 17, T = 20$), so QT is the odd one out.

2.1.16 Inequalities (reasoning)

What it is: Deducing the relationship between persons or letters from chains using the signs $>$, $<$, $=$, $\geq$, $\leq$. Sometimes the signs are replaced by codes such as @ and #.

How to identify it

- "Statement: $A > B \geq C = D < E$. Conclusions: I. $A > D$ II. $C < E$ ".
- Coded symbols such as "A @ B means A is greater than B".

How to solve it

- Decode the symbols into the real signs first.
- A conclusion between two letters is possible only if the chain between them points in one direction the whole way.
- Combining signs: $>$ with $>$ gives $>$; $>$ with $\geq$ gives $>$; $\geq$ with $\geq$ gives $\geq$; $=$ with any sign keeps that sign. Mixed directions ($<$ then $>$) give no conclusion.
- If neither of two conclusions about the same pair is certain, but together they cover every possibility (for example " $A > B$ " and " $A \leq B$ ", or " $A = B$ " and " $A \neq B$ "), then "either I or II follows".

Example 1. Statement: $P \geq Q > R = S < T$. Conclusions: I. $P > S$. II. $T > R$.

Solution. $P \geq Q > R = S$, so $P > S$ ($\geq$ followed by $>$ gives $>$). Conclusion I follows.

$R = S < T$, so $R < T$. Conclusion II follows.

Both follow.

Example 2. Statement: $A < B \leq C > D$. Conclusions: I. $A < C$. II. $B > D$.

Solution. $A < B \leq C$, so $A < C$. Conclusion I follows.

For B and D the chain goes $B \leq C$ and then $C > D$, which changes direction, so no relation can be fixed.

Only I follows.

2.1.17 Input-output

What it is: A machine rearranges a list of numbers or words by a fixed rule, one step at a time. You find the rule and predict a later step.

How to identify it

- "Input: ... Step I: ... Step II: ... Find Step V" or "which step is the final arrangement".

How to solve it

- Compare the input with Step I and Step I with Step II to see exactly what moved.
- Typical rules: smallest or largest item moves to one end each step; items sorted alphabetically one at a time; numbers rearranged by digit sum; alternate items swapped.
- Apply the same rule again to get later steps. The last step is the fully arranged list, usually after (number of items – 1) steps.

Example 1. Input: 52 17 83 29 41. Step I: 17 52 83 29 41. Step II: 17 29 52 83 41. Find the rule and Step III.

Solution. Each step brings the next smallest remaining number to the left and the other numbers keep their order. Step III brings 41 forward: 17 29 41 52 83.

Example 2. Input: tree sky moon star. Step I: moon tree sky star. Step II: moon sky tree star. Find Step III.

Solution. In alphabetical order the words are moon, sky, star, tree. Step I moves the first of these (moon) to the front. Step II moves the next one (sky) into second place. The remaining words keep their order. Step III moves star into third place: moon sky star tree.

2.1.18 Mathematical operations

What it is: The usual signs (+, –, ×, ÷) are given new meanings, or a new operation is defined by a rule. You must evaluate an expression with the new meanings.

How to identify it

- "If + means ×, × means –, ... then find the value of..."
- "If $a * b = 2a + b$, find $3 * 4$ ".

How to solve it

- Rewrite the whole expression, replacing each symbol with the operation it stands for. Do the replacement once, for every symbol at the same time.
- Then use BODMAS on the new expression.
- For a defined operation, substitute the numbers into the rule. For nested brackets, evaluate the inner one first.

Example 1. If + means ×, × means –, – means ÷ and ÷ means +, find $12 + 3 \times 4 - 2 \div 5$.

Solution. Replace the symbols: $12 \times 3 - 4 \div 2 + 5$.

Divide first: $4 \div 2 = 2$. Multiply: $12 \times 3 = 36$.

$36 - 2 + 5 = 39$.

Example 2. If $a * b = 2a + b$, find $3 * 4$ and $(2 * 3) * 1$.

Solution. $3 * 4 = 2 \times 3 + 4 = 10$.

$2 * 3 = 2 \times 2 + 3 = 7$, and then $7 * 1 = 2 \times 7 + 1 = 15$.

So the values are 10 and 15.

2.1.19 Alphanumeric series

What it is: A mixed sequence of letters, digits and symbols, with questions about counting or finding elements by position.

How to identify it

- "How many such 7s are immediately preceded by a vowel and followed by a symbol?"
- "Which element is fifth to the right of E?"

How to solve it

- Copy the series and number each position from the left.
- Treat each condition one at a time and mark the matches. "Preceded by" is the element on the left, "followed by" is the element on the right.
- For "from the right" questions, count from the right end instead.

Example 1. Series: 5 A 7 # 3 E 7 4 9 U 7 @ 2 I 4. How many 7s are immediately preceded by a vowel and immediately followed by a symbol?

Solution. Positions: 1:5, 2:A, 3:7, 4:#, 5:3, 6:E, 7:7, 8:4, 9:9, 10:U, 11:7, 12:@, 13:2, 14:I, 15:4.

The 7 at position 3 is preceded by A and followed by # (counts). The 7 at position 7 is preceded by E but followed by 4 (does not count). The 7 at position 11 is preceded by U and followed by @ (counts).

The answer is 2.

Example 2. In the same series, how many vowels are immediately followed by a digit? Which element is fifth to the right of E?

Solution. A is followed by 7, E by 7, U by 7 and I by 4. All four vowels are followed by a digit, so the count is 4.

E is at position 6. Fifth to its right is position 11, which is 7.

2.1.20 Truth & lie puzzles

What it is: People make statements; some always tell the truth and some always lie. You find out who is who.

How to identify it

- "A says B is lying", "exactly one of them is telling the truth", "who broke the glass".

How to solve it

- Assume one person is truthful and follow what that implies. If it leads to a contradiction, that assumption is wrong.
- A truthful person's statements are all true. A liar's statements are all false, so the opposite of what a liar says is true.
- Use any count given ("exactly two speak the truth") to test each possible culprit in turn.

Example 1. A says "B is lying". B says "C is lying". C says "A and B are both lying". Who is telling the truth?

Solution. Suppose C is truthful. Then A and B are both lying. But if A is lying, then "B is lying" is false, so B is truthful. That contradicts B lying. So C cannot be truthful, and C lies.

Since C lies, "A and B are both lying" is false, so at least one of A and B is truthful.

Suppose A is truthful. Then B lies, so "C is lying" is false and C is truthful. That contradicts the previous step, so A is not truthful.

So B is truthful. Check: "C is lying" is true. A says "B is lying", which is false, so A lies. All three statements fit.

Only B tells the truth. A and C lie.

Example 2. A glass was broken by Ram, Shyam or Mohan. Ram says "Shyam did it". Shyam says "Mohan did it".

Mohan says "I did not do it". Exactly two of them speak the truth. Who broke the glass?

Solution. If Ram did it: Ram false, Shyam false, Mohan true. One truth, which is not two.

If Shyam did it: Ram true, Shyam false, Mohan true. Two truths, which matches.

If Mohan did it: Ram false, Shyam true, Mohan false. One truth.

Shyam broke the glass.

2.1.21 Critical reasoning

What it is: Reading a short argument and judging it: finding the assumption, the flaw, or what strengthens or weakens it.

How to identify it

- "Which of the following, if true, weakens (or strengthens) the argument?", "which is an assumption", "what is the flaw".

How to solve it

- Split the passage into the premise (reason) and the conclusion (what is claimed). Words such as "therefore", "so", "hence" mark the conclusion.
- To strengthen: pick the option that supports the link between premise and conclusion, or rules out another explanation.
- To weaken: pick the option that gives another explanation or shows the link fails.
- Assumption: the missing link that must be true. Negate an option; if the argument collapses, it is the assumption.
- Avoid options that go beyond the passage or are too extreme.

Example 1. Argument: "Sales of Brand X rose after it hired a famous actor for its advertisements, so the actor's advertisements caused the rise." Which statement weakens this most? (a) The actor is very popular. (b) In the same month a rival brand closed down, and its customers moved to Brand X.

Solution. Option (b) gives another reason for the rise in sales, so the actor may not be the cause. It weakens the argument.

Option (a) fits the argument and does not weaken it.

Example 2. Argument: "Air quality in the city improved after the car-free day, so car-free days improve air quality." Which statement strengthens this most? (a) Many people enjoyed the car-free day. (b) On comparable days with no special measures, air quality did not change.

Solution. Option (b) rules out the idea that air quality would have improved anyway, so it supports the claim. Option (a) is about enjoyment, which is irrelevant to air quality.

2.1.22 Decision making

What it is: Applying a set of rules or conditions to candidates (for a job, loan or admission) to decide the outcome for each case.

How to identify it

- "Conditions for selection are... Decide whether the candidate should be selected, rejected or referred".

How to solve it

- Turn the rules into a checklist.
- Test each candidate against every rule. If a compulsory rule fails, stop and look for the stated exception.
- Use only the rules given; do not bring in outside opinions.

Example 1. Rules: a candidate must (i) be a graduate, (ii) be aged 21 to 30, and (iii) have at least 2 years of experience. If all are met, select. If only (iii) is not met but the candidate is a post-graduate, refer to HR. In all other cases, reject. Candidate: graduate, 25 years, 1 year of experience, not a post-graduate. Decision?

Solution. The candidate meets (i) and (ii) but fails (iii).

He is not a post-graduate, so the exception does not apply.

Decision: reject.

Example 2. Same rules. Candidate: post-graduate, 28 years, 1 year of experience. Decision?

Solution. This candidate meets (i) and (ii) and fails only (iii), and is a post-graduate.

The exception applies.

Decision: refer to HR.

2.1.23 Data sufficiency (reasoning)

What it is: A reasoning question is followed by statements. You decide whether the statements give enough information to answer it.

How to identify it

- "Who is the tallest among five?", "How is M related to N?" followed by statements I and II, with options like "both are needed".

How to solve it

- Try to answer from statement I alone, then II alone, then both.
- Sufficient means exactly one definite answer. Two possible answers means not sufficient.
- Draw the order or family tree each time, so you can see if a unique answer emerges.

Example 1. Who is the heaviest among P, Q and R? I. P is heavier than Q. II. R is lighter than P.

Solution. I alone gives $P > Q$ only; R is unknown.

II alone gives $P > R$ only; Q is unknown.

Together, P is heavier than both Q and R, so P is the heaviest. Both statements are needed.

Example 2. How is M related to N? I. M is the sister of N's father. II. M is a daughter of N's grandfather.

Solution. I alone: M is N's father's sister, so M is N's paternal aunt. Sufficient.

II alone: if the grandfather is N's father's father, M is N's aunt; if he is N's mother's father, M could be N's mother or an aunt. More than one answer, so not sufficient.

Statement I alone is sufficient.

2.2 Non-Verbal Reasoning

2.2.1 Series (figures)

What it is: A row of figures that change step by step by rotating, adding or removing parts, or changing shading. You choose the next figure.

How to identify it

- "Find the next figure in the series", figures such as arrows, shapes with dots, shapes that grow.

How to solve it

- Follow one feature at a time: rotation, position of a dot or line, number of parts, shading.
- Rotation series: find the angle per step (45° or 90° clockwise or anticlockwise).
- If the series looks irregular, split it into two interleaved series (odd and even steps).
- Count sides, lines or dots; they often follow a number pattern.

Example 1. An arrow points up in the first figure, right in the second and down in the third. Which direction does it point in the fourth figure?

Solution. The arrow turns 90° clockwise at each step: up, right, down.
The next direction is left.

Example 2. A triangle has a dot on one vertex. The dot is on the top vertex, then the bottom-right vertex, then the bottom-left vertex. Where is the dot in the next figure?

Solution. The dot moves one vertex clockwise each time (top $\rightarrow$ bottom-right $\rightarrow$ bottom-left).
The next move goes back to the top vertex.

2.2.2 Analogy (figures)

What it is: A pair of figures shows a change. You apply the same change to a third figure to get the fourth.

How to identify it

- "Figure A is to figure B as figure C is to...?"

How to solve it

- Describe the change from A to B in words: turned, flipped, shaded, enlarged, a part added or removed, one shape replaced by another.
- Apply exactly that change to C.
- Check the answer options one feature at a time, such as shape, direction, shading.

Example 1. A is a white square and B is a black square. C is a white circle. What is D?

Solution. The change from A to B is "the shape becomes shaded black".
D is a black circle.

Example 2. A is an arrow pointing up and B is an arrow pointing right. C is an arrow pointing left. What is D?

Solution. The change from A to B is a 90° clockwise turn.
Turning C (left) by 90° clockwise gives up.
D is an arrow pointing up.

2.2.3 Odd one out (figures)

What it is: Out of several figures, one breaks a rule that all the others obey.

How to identify it

- "Find the odd figure out" among shapes, letters or patterns.

How to solve it

- Compare the figures for: number of sides or lines, open or closed, shaded or not, symmetry, number of intersections, direction of a part.
- Find a rule true for all but one. The exception is the answer.
- If two rules seem to fit, choose the one that separates exactly one figure.

Example 1. Four figures: a triangle, a square, a pentagon and a circle. Which is the odd one out?

Solution. The triangle, square and pentagon are made of straight sides (polygons).
The circle has no straight side, so the circle is the odd one out.

Example 2. Four letters: A, H, M, F. Which is the odd one out?

Solution. A, H and M each have a vertical line of symmetry (left and right halves match).
F does not, so F is the odd one out.

2.2.4 Mirror & water images

What it is: A mirror image is a left-to-right reversal. A water image is an upside-down (top-to-bottom) reversal, as seen in the water below the object.

How to identify it

- "Mirror image of the figure", "water image of", "the time shown in a mirror".

How to solve it

- Mirror held at the right or left: swap left and right. The first letter of a word appears last and every letter is reversed horizontally.
- Mirror at the top or bottom, or water image: swap top and bottom. Letters turn upside down.
- Clock face: mirror image time = $12:00 - \text{actual time}$ (or $11:60 - \text{time}$). Water image time = $6:00 - \text{actual time}$.
- Letters that look the same in a left-right mirror: A, H, I, M, O, T, U, V, W, X, Y.

Example 1. What time will a mirror show if the clock shows 4:20?

Solution. Mirror image time = $11:60 - 4:20$.

$11:60 - 4:20 = 7:40$.

Example 2. What is the water image of the time 4:20?

Solution. Water image time = $6:00 - 4:20$.

$6:00 - 4:20 = 1:40$.

2.2.5 Paper folding & cutting

What it is: A sheet of paper is folded one or more times, a hole is punched or a shape is cut, and the sheet is unfolded. You predict the pattern.

How to identify it

- "A square paper is folded as shown and a hole is punched. How will it look when opened?"

How to solve it

- Unfold in the reverse order: undo the last fold first.
- At each unfolding, copy every cut or hole across the fold line to the other side (mirror image).
- A hole through n layers makes n holes. Two folds in half give 4 layers, so one punch makes 4 holes (if the hole is not on a fold line).
- Holes on a fold line become half holes that join into one full hole.

Example 1. A square paper is folded in half once and a single small hole is punched near the fold, but not on it. How many holes appear after unfolding and where?

Solution. The paper had 2 layers, so there are 2 holes.

They are mirror images across the fold line, at equal distances on either side.

Example 2. A square paper is folded in half twice (so it is a quarter of its size) and one hole is punched in the middle of the folded square. How many holes appear after unfolding?

Solution. The paper had 4 layers, so there are 4 holes.

There is one in each quarter of the sheet, placed symmetrically about the centre lines.

2.2.6 Embedded figures

What it is: A simple figure is hidden inside a more complex figure. You find which answer figure is part of it.

How to identify it

- "Which of the answer figures is embedded in the question figure?"

How to solve it

- Study the simple figure for a distinctive feature, such as its longest line or an unusual angle.
- Look for that feature in the large figure, and trace the simple figure over it in the same size and the same orientation (no turning).
- Eliminate options that have a line not present in the large figure.

Example 1. The question figure is a square with both diagonals drawn. Which is embedded: (a) a triangle (b) a smaller square (c) a circle (d) a pentagon?

Solution. Two half-diagonals and one side form a triangle, so (a) is embedded.

No smaller square, circle or pentagon can be traced.

Answer: (a).

Example 2. A square has both diagonals drawn. Which of these is NOT embedded: (a) a triangle (b) a smaller square (c) the letter X (d) the letter V?

Solution. (a) Triangle: yes. (c) The letter X is the two diagonals. (d) V is two half-diagonals meeting at the centre. A smaller square cannot be traced anywhere.

Answer: (b).

2.2.7 Pattern completion

What it is: A design has a missing part. You choose the piece that completes it.

How to identify it

- "Which of the answer pieces fits in the blank space to complete the pattern?"

How to solve it

- Follow every line that enters the gap and check which option continues it without a break.
- Check the repeat or symmetry of the design (what appears in the other rows and columns).
- Match colour, shading, line spacing and angle.
- Remove options one by one that clearly break a line, spacing or colour.

Example 1. A 4×4 checkerboard of black and white tiles has one tile missing. The tiles to its left and above it are black. What colour is the missing tile?

Solution. Tiles in a checkerboard alternate colours with each neighbour.

Its neighbours are black, so the missing tile is white.

Example 2. A design has parallel stripes sloping upwards to the right, with equal gaps. The options are: (a) stripes sloping down to the right, (b) up-right stripes with double the gap, (c) up-right stripes with the same gap, meeting the edges where the neighbouring stripes do, (d) up-right stripes, shifted so the lines break at the edges. Which completes it?

Solution. The slope must match, which removes (a). The gap must match, which removes (b). The stripes must join the neighbouring stripes at the edges, which removes (d).

Answer: (c).

2.2.8 Counting figures

What it is: Counting all the triangles, squares, rectangles or lines hidden in a figure.

How to identify it

- "How many triangles are there in the figure?", "how many squares", "count the straight lines".

How to solve it

- Name the points with letters, then count in a fixed order: smallest pieces first, then pieces made of 2, then 3, and so on.
- A triangle with lines drawn from the apex to points on the base, dividing the base into n parts: number of triangles = $n(n + 1)/2$.
- An $n \times n$ grid: squares = $n(n + 1)(2n + 1)/6$, rectangles (including squares) = $[n(n + 1)/2]^2$.
- Do not count the same piece twice: tick each one as you list it.

Example 1. How many squares and how many rectangles (including squares) are there in a 3×3 grid?

Solution. Squares = $3^2 + 2^2 + 1^2 = 9 + 4 + 1 = 14$.

Rectangles = $(3 \times 4/2)^2 = 6^2 = 36$.

Example 2. In triangle ABC, 4 lines are drawn from A to points on BC, which divides BC into 5 parts. How many triangles are there?

Solution. Number of triangles = $n(n + 1)/2$ with $n = 5$.

$5 \times 6/2 = 15$.

2.2.9 Cubes & dice

What it is: Questions on a painted cube cut into smaller cubes, and on finding the face opposite another face of a cube or dice from several views.

How to identify it

- "A cube painted on all faces is cut into 64 small cubes. How many have exactly two faces painted?"
- "Two positions of a dice are shown. What is opposite to...?"

How to solve it

- Cube of side n cut into n^3 unit cubes: 3 painted faces = 8 (corners); 2 painted faces = $12(n - 2)$; 1 painted face = $6(n - 2)^2$; no painted face = $(n - 2)^3$.
- Dice views: faces seen together in one view are adjacent, so they cannot be opposite each other.
- If two views share two faces, the remaining face in each view are opposite each other.
- A face never seen with another one is a candidate for being opposite to it.

Example 1. A cube painted on all faces is cut into 64 equal small cubes. How many have exactly two faces painted?

Solution. $64 = 4^3$, so $n = 4$.

Two painted faces = $12 \times (4 - 2) = 24$.

Check: $8 + 24 + 24 + 8 = 64$ (three-face, two-face, one-face, no-face cubes).

Example 2. A cube has faces marked A to F. In one view A, B and C are visible together. In another view A, C and D are visible together. Which face is opposite B?

Solution. Faces A and C appear in both views, so they share an edge.

B and D are the two faces that touch both A and C, which are on opposite ends of that edge, so B and D are opposite.

The face opposite B is D.

2.2.10 Figure matrix

What it is: A grid (usually 3×3) of figures follows a rule along rows or columns. One cell is missing and you find it.

How to identify it

- "Find the figure that should replace the question mark in the matrix."

How to solve it

- Test the rows first. Common rules: each row has one of each shape, the third cell is the sum or difference of the first two, shading rotates.
- If rows give no rule, test the columns.
- Examine one feature at a time: shape, count, shading, orientation.
- Confirm your guess against the completed rows or columns.

Example 1. A matrix shows the number of dots. Row 1: 2, 3, 5. Row 2: 4, 1, 5. Row 3: 6, 2, ? What is the missing number?

Solution. In rows 1 and 2 the third cell is the sum of the first two ($2 + 3 = 5$ and $4 + 1 = 5$).

Row 3: $6 + 2 = 8$.

Example 2. Row 1: circle, square, triangle. Row 2: square, triangle, circle. Row 3: triangle, circle, ? What is the missing shape?

Solution. Each row contains a circle, a square and a triangle exactly once.

Row 3 has a triangle and a circle, so the missing shape is a square.

Part 3: Verbal Ability

English usage and comprehension. Grammar questions reward rules; vocabulary rewards daily reading; passages reward careful, calm reading.

3.1 Grammar

3.1.1 Parts of speech

What it is: The eight word classes: noun (names), pronoun (replaces a noun), verb (action or state), adjective (describes a noun), adverb (describes a verb, adjective or adverb), preposition (shows relation), conjunction (joins), interjection (exclamation).

How to identify it

- "Identify the part of speech of the underlined word."
- Sentences where an adjective is used in place of an adverb, or the reverse (good/well, quick/quickly).

How to solve it

- Decide by the job the word does in the sentence, not by how it looks. The same word can change class ("run" is a verb in "I run" and a noun in "a long run").
- Adverbs usually answer how, when, where or how much and often end in -ly, but not always (fast, hard, late).
- After linking verbs (is, seems, feels, looks, becomes) use an adjective. After action verbs use an adverb.
- Remember: "good" is an adjective and "well" is the adverb.

Example 1. Identify the part of speech of "fast" in (a) "He drives a fast car." and (b) "He drives fast."

Solution. (a) "Fast" describes the noun "car", so it is an adjective.

(b) "Fast" describes the verb "drives", so it is an adverb.

Example 2. Choose the correct word: "She sings very (beautiful / beautifully)."

Solution. "Sings" is an action verb, so it needs an adverb to describe how she sings.

Answer: beautifully.

3.1.2 Tenses

What it is: Verb forms that show when something happens and whether it is complete: simple, continuous, perfect and perfect continuous, each in the past, present and future.

How to identify it

- Time words: since, for, yesterday, ago, already, just, by next month, by the time.
- Blanks where the verb is given in brackets.

How to solve it

- Present simple: habits and facts. Present continuous: happening now (is/am/are + -ing).
- Present perfect (has/have + V3): finished action linked to now, used with since, for, already, just, yet.
- Past simple: finished action at a definite past time (yesterday, in 2010, ago).
- Past perfect (had + V3): the earlier of two past actions. Future perfect (will have + V3): completed before a future time ("by then").
- "Since" goes with a point in time (since 2018). "For" goes with a length of time (for six years).

Example 1. Fill the blank: "I (live) in Hyderabad since 2018."

Solution. "Since" with a starting point and an action that continues up to now needs the present perfect.

Answer: have lived.

Example 2. Fill the blank: "By the time we reached the station, the train (leave)."

Solution. Two past actions: the train left first, then we arrived. The earlier action takes the past perfect.

Answer: had left.

3.1.3 Subject-verb agreement

What it is: The verb must agree with its subject in number (singular or plural) and person.

How to identify it

- A long phrase sits between the subject and the verb.
- Words like each, every, either, neither, a number of, along with, as well as, or collective nouns (team, committee).

How to solve it

- Find the real subject. Ignore phrases that begin with of, along with, as well as, together with, in addition to.
- Each, every, either, neither, one of the... take a singular verb.
- Either...or, neither...nor, not only...but also: the verb agrees with the nearer subject.
- "A number of" takes a plural verb. "The number of" takes a singular verb.
- Uncountable nouns (information, advice, furniture, news) take a singular verb. A collective noun acting as one unit is singular.

Example 1. Choose the correct verb: "The quality of these mangoes (is / are) poor."

Solution. The subject is "quality", not "mangoes". "Of these mangoes" is only a phrase.

Answer: is.

Example 2. Choose the correct verb: "Neither the manager nor his assistants (was / were) present."

Solution. With neither...nor, the verb agrees with the nearer subject, "assistants", which is plural.

Answer: were.

3.1.4 Articles

What it is: The words a, an and the. "A" and "an" point to any one item; "the" points to a specific or known item.

How to identify it

- Blanks that ask for a, an, the or no article.
- Sentences with errors such as "a hour" or "an university".

How to solve it

- Use "an" before a vowel sound and "a" before a consonant sound, whatever the spelling: an hour, an MBA, a university, a European.
- Use "the" for something specific or already mentioned, for unique things (the sun), superlatives (the best), rivers, oceans, mountain ranges and "the + adjective" for a group (the poor).
- Use no article before plural or uncountable nouns in general statements, languages, meals, and most names of people and places.

Example 1. Fill in the blanks: "He is (a / an) honest man and (a / an) MBA graduate."

Solution. "Honest" starts with a silent h, so it begins with a vowel sound: an honest man.

"MBA" is pronounced "em-bee-ay", a vowel sound: an MBA graduate.

Example 2. Fill in the blanks: "(The / -) Ganga is (the / -) longest river in India."

Solution. Rivers take "the", and a superlative takes "the".

Answer: The Ganga is the longest river in India.

3.1.5 Prepositions

What it is: Words that show the relation of a noun or pronoun to another word, in time, place or direction (in, on, at, by, from, to, with...).

How to identify it

- Blanks with a choice of at/on/in/by, or a word that needs a fixed preposition (good at, afraid of).

How to solve it

- Time: "at" for clock times, "on" for days and dates, "in" for months, years, seasons and long periods.
- Place: "at" a point, "in" an enclosed space or large area, "on" a surface.
- "Between" is for two, "among" for more than two.
- Learn fixed pairs: good at, afraid of, interested in, fond of, married to, different from, capable of, depend on, congratulate on, prefer X to Y, senior or junior to.

Example 1. Fill in the blanks: "The meeting is at 5 p.m. (at / on / in) Monday (at / on / in) August."

Solution. Days take "on". Months take "in".

Answer: on Monday, in August.

Example 2. Fill in the blanks: "She is good (at / in / of) mathematics and fond (of / on / at) puzzles."

Solution. The fixed phrases are "good at" and "fond of".

Answer: at, of.

3.1.6 Pronouns

What it is: Words used instead of nouns: personal (I, me, he, him), possessive (his, mine), reflexive (himself), relative (who, whom, which, that), demonstrative (this, that) and indefinite (someone, each).

How to identify it

- Blanks choosing between I/me, he/him, who/whom.
- "Each other" or "one another", "its" or "it's", or a pronoun that does not match its noun.

How to solve it

- Subject forms (I, he, she, we, they) go before a verb. Object forms (me, him, her, us, them) go after a verb or preposition: "between you and me".
- "Who" is for the subject. "Whom" is for the object. Test by replacing with he/him: if "him" fits, use "whom".
- A pronoun must match the noun it refers to in number and gender.
- "Each other" is for two. "One another" is for more than two. "Its" is possessive; "it's" means "it is".

Example 1. Choose the correct pronoun: "He gave the book to (I / me / myself)."

Solution. "To" is a preposition and needs the object form.

Answer: me.

Example 2. Choose the correct pronoun: "The man (who / whom) I met yesterday is a doctor."

Solution. "I met him": the man is the object of "met".

Answer: whom.

3.1.7 Conjunctions

What it is: Words that join words, phrases or clauses and show how the ideas are related.

How to identify it

- Blanks with although/because/unless/but, pairs such as neither...nor, no sooner...than.
- A sentence with two joining words for the same idea ("Although... but...").

How to solve it

- Coordinating: and, but, or, so, yet. Subordinating: although, though, because, since, unless, while, whereas, until.
- "Although" or "though" already shows contrast, so do not add "but" or "yet" in the same sentence.
- "Unless" means "if not", so do not add another "not".
- Correlatives come in fixed pairs: either...or, neither...nor, both...and, not only...but also, hardly/scarcely...when, no sooner...than.

Example 1. Fill the blank: "(Although / Because) he was tired, he finished the work."

Solution. The two ideas are in contrast (tired, yet finished).

Answer: Although.

Example 2. Fill the blank: "No sooner had he reached the station (than / when) the train left."

Solution. "No sooner" is always paired with "than". ("Hardly" is paired with "when".)

Answer: than.

3.1.8 Active & passive voice

What it is: In the active voice the subject does the action. In the passive voice the subject receives the action.

How to identify it

- "Change the following sentence into the passive voice" (or active).
- Verb forms such as "was written", "has been done", "is being prepared".

How to solve it

- Make the object the new subject. Use a form of "be" in the same tense, then the past participle (V3). Add "by + doer" only if it matters.
- Keep the tense: is/am/are + V3 (present), was/were + V3 (past), will be + V3 (future), has/have been + V3 (present perfect), is/was being + V3 (continuous), can/must be + V3 (modals).
- Change pronouns: I → me, he → him, they → them.

Example 1. Change into the passive: "The chef cooked the meal."

Solution. Object "the meal" becomes the subject. Past tense: was + cooked.
The meal was cooked by the chef.

Example 2. Change into the passive: "They have completed the project."

Solution. Present perfect becomes "has been + V3".
The project has been completed by them.

3.1.9 Direct & indirect speech

What it is: Direct speech repeats the speaker's exact words in quotation marks. Indirect (reported) speech gives their meaning in your own words.

How to identify it

- "Change the following into indirect speech", sentences with quotation marks and "said" or "asked".

How to solve it

- Remove the quotation marks. Use "that" for statements, "if" or "whether" for yes/no questions, the question word for wh-questions (in statement order), and "to" for commands.
- If the reporting verb is in the past, move the tense back: am/is → was, do/does → did, will → would, can → could, past simple → past perfect.
- Change pronouns by meaning. Time and place words: now → then, today → that day, tomorrow → the next day, yesterday → the previous day, here → there, this → that.
- Universal truths and habits do not change tense.

Example 1. Change into indirect speech: He said, "I am going to the market tomorrow."

Solution. "Said" is past, so "am going" becomes "was going". "I" becomes "he". "Tomorrow" becomes "the next day".
He said that he was going to the market the next day.

Example 2. Change into indirect speech: She asked, "Where do you live?"

Solution. A wh-question: keep the question word, use statement order, move the tense back.
She asked where I lived.

3.1.10 Modifiers

What it is: A modifier is a word or phrase that describes another word. Errors arise when it is misplaced or when it is left without anything to describe (a dangling modifier).

How to identify it

- A sentence that sounds odd because the description seems to apply to the wrong noun ("Walking down the street, a car hit Ravi.").
- "Only", "almost", "just" placed in the wrong position.

How to solve it

- Place the modifier next to the word it describes.
- When a sentence opens with a phrase such as "Walking down the road,", the very next noun must be the person or thing doing that action.
- "Only", "almost", "just" go directly before the word they limit.
- Fix by rewriting: either change the main clause so the right noun follows, or turn the phrase into a clause with its own subject.

Example 1. Correct the sentence: "Walking down the street, a car hit Ravi."

Solution. As written, the car was walking. The person walking was Ravi.

Correct: "Walking down the street, Ravi was hit by a car." or "While Ravi was walking down the street, a car hit him."

Example 2. Which sentence means that she drove close to 200 km? (a) She almost drove 200 km. (b) She drove almost 200 km.

Solution. In (a), "almost drove" suggests she nearly drove but did not. In (b), "almost" limits "200 km".

Answer: (b).

3.1.11 Parallelism

What it is: Items in a list, or two parts of a comparison, must have the same grammatical form.

How to identify it

- Lists joined by and/or, "not only...but also", "either...or", comparisons using than or as.
- Mixed forms such as "swimming, to read and cooking".

How to solve it

- Check every item in the list: all nouns, all -ing forms, or all infinitives.
- After a correlative pair (not only...but also), the same structure must follow each part.
- Compare like with like: "The population of Mumbai is larger than that of Delhi" (not "than Delhi").

Example 1. Correct the sentence: "She likes swimming, to read and cooking."

Solution. The list mixes -ing forms with an infinitive.

Correct: "She likes swimming, reading and cooking."

Example 2. Correct the sentence: "The population of Mumbai is larger than Delhi."

Solution. It compares a population with a city.

Correct: "The population of Mumbai is larger than that of Delhi."

3.1.12 Error spotting

What it is: A sentence is divided into parts. You find the part with a grammatical error, or say there is none.

How to identify it

- A sentence split into (A), (B), (C), (D) with an option "No error".

How to solve it

- Read the full sentence, then test each part against the standard checks: subject-verb agreement, tense, article, preposition, pronoun case, adjective versus adverb, word order, comparison, conjunction pairs.
- Watch the common traps: each/every with a plural verb, "less" for countable nouns (use "fewer"), "since/for", "among/between", "senior to" (not "than").
- If every part passes the checks, choose "No error".

Example 1. Find the part with the error: (A) Each of the students / (B) have submitted / (C) their assignment / (D) on time.

Solution. "Each" is singular and needs a singular verb.

Error in (B): it should be "has submitted".

Example 2. Find the part with the error: (A) He is senior / (B) than me / (C) by five years. / (D) No error.

Solution. "Senior", "junior" and "prefer" take "to", not "than".

Error in (B): it should be "to me".

3.1.13 Sentence correction

What it is: A sentence has an underlined part. You choose the best replacement, or "no improvement".

How to identify it

- "Select the option that improves the underlined part", with the first option usually repeating the original.

How to solve it

- First find what is wrong in the original: agreement, tense, parallelism, modifier, idiom, wordiness.
- Remove options that keep the error or create a new one.
- Prefer the shortest option that is clear, correct and parallel.
- Choose "no improvement" only when you cannot find an error.

Example 1. Improve the sentence: "The committee have announced its decision." (a) No improvement (b) has announced its decision (c) have announced their decision

Solution. The committee acts as one unit, so it takes a singular verb, matching "its".

Answer: (b).

Example 2. Improve the sentence: "He is one of those who always helps others." (a) No improvement (b) who always help others (c) who is always helping others

Solution. In "one of those who...", the word "who" refers to the plural "those", so the verb must be plural.

Answer: (b).

3.2 Vocabulary

3.2.1 Synonyms & antonyms

What it is: Synonyms are words with similar meanings. Antonyms are words with opposite meanings.

How to identify it

- "Choose the word closest in meaning to..." or "most opposite in meaning to...", usually with the target word in capitals.

How to solve it

- Read the sentence for context and decide whether the word is positive or negative in tone.
- Break unfamiliar words using roots, prefixes and suffixes (un-, in-, dis-, mal- often give a negative sense; bene- means good, mal- means bad).
- Remove options with the wrong tone first.
- For antonyms, do not pick a synonym by mistake; re-read the question before you answer.

Example 1. Choose the synonym of ABUNDANT: (a) scarce (b) plentiful (c) rare (d) little

Solution. "Abundant" means present in large quantity.

Answer: (b) plentiful.

Example 2. Choose the antonym of BENEVOLENT: (a) kind (b) generous (c) malevolent (d) helpful

Solution. "Bene-" means good, so "benevolent" means wishing good to others. Its opposite uses "male-", meaning bad.

Answer: (c) malevolent.

3.2.2 One-word substitution

What it is: Replacing a long phrase with a single exact word.

How to identify it

- "A person who...", "a place where...", "the study of...", "fear of...", "government by..."

How to solve it

- Learn the common roots and endings: -cracy (rule), -ology (study), -phile (lover), -phobia (fear), -cide (killing), anthropo- (man), bio- (life), omni- (all).
- Keep a list of frequent examples: astronomer, philanthropist, atheist, omnivorous, misogynist, bibliophile, pedestrian.
- Test the word by saying the meaning in full: if it fits the phrase exactly, it is correct.

Example 1. Give one word for "a person who loves books".

Solution. "Biblio-" refers to books and "-phile" means lover.

Answer: bibliophile.

Example 2. Give one word for (a) "government by the people" and (b) "fear of closed spaces".

Solution. (a) "Demos" means people and "-cracy" means rule: democracy.

(b) "-phobia" means fear. For closed spaces: claustrophobia.

3.2.3 Idioms & phrases

What it is: Expressions whose meaning is not the same as the meaning of their separate words.

How to identify it

- "Choose the meaning of the idiom", "the phrase 'a blessing in disguise' means..."

How to solve it

- Never translate word by word. Use the context of the sentence to guess the meaning and eliminate literal options.
- Learn idioms in groups with a story or image, and reuse them in your own sentences.
- Common ones: bite the bullet (face a difficulty bravely), once in a blue moon (very rarely), a blessing in disguise (a good thing that seemed bad at first), break the ice (start a conversation), hit the nail on the head (be exactly right).

Example 1. What does "to bite the bullet" mean?

Solution. It means to face a painful or difficult situation with courage instead of avoiding it.

Example 2. What do "a blessing in disguise" and "once in a blue moon" mean?

Solution. "A blessing in disguise" is something that looks bad at first but turns out to be good.

"Once in a blue moon" means very rarely.

3.2.4 Spelling check

What it is: Picking the correctly spelt word from several options.

How to identify it

- "Choose the correctly spelt word" or "find the misspelt word".

How to solve it

- Learn the main rules: "i before e, except after c" (receive, believe); double letters (accommodation, occurrence, embarrass); silent letters (knowledge, colonel, island).
- Split the word into syllables and picture it: sep-a-rate, nec-es-sary.
- Learn common troublemakers and revise them regularly.
- Words ending in -able/-ible: learn them as a list (readable, possible).

Example 1. Choose the correct spelling: (a) Accomodation (b) Accommodation (c) Acommodation (d) Acomodation

Solution. The word has a double c and a double m.

Answer: (b).

Example 2. Choose the correct spelling: (a) Recieve (b) Receive (c) Receeve (d) Receve

Solution. "I before e, except after c": after the c, the order is e then i.

Answer: (b).

3.2.5 Homonyms & confusables

What it is: Words that sound or look alike but differ in meaning or use, such as affect and effect.

How to identify it

- Blanks that need a choice between affect/effect, principal/principle, stationary/stationery, advice/advise, complement/compliment.

How to solve it

- Decide the part of speech the blank needs (noun or verb), then pick the matching word.
- affect (verb) and effect (noun). advice (noun) and advise (verb). principal (head, chief) and principle (rule). stationary (not moving) and stationery (paper and pens). complement (completes) and compliment (praise). lose (misplace) and loose (not tight).
- their (belongs to them), there (place), they're (they are). its (possessive), it's (it is). than (comparison), then (time).

Example 1. Choose the correct word: "The new rule will (affect / effect) everyone."

Solution. A verb is needed here.

Answer: affect.

Example 2. Choose the correct words: "The (principal / principle) announced a holiday, and she believes in the (principal / principle) of honesty."

Solution. The head of a school is the principal. A rule of conduct is a principle.
Answer: principal, principle.

3.2.6 Word analogy

What it is: Pairs of words are linked by a relationship. You find the word that completes a second pair with the same relationship.

How to identify it

- "Hot : Cold :: Day : ?", "Pilot : Aeroplane :: Driver : ?"

How to solve it

- State the link between the first pair in one sentence (opposite, worker and tool, part and whole, degree, cause and effect).
- Make the same sentence for each option and keep the one that fits.
- Keep the order the same: if the first word is the person, the second must be the thing.

Example 1. Hot : Cold :: Day : ? (Sun, Night, Light, Morning)

Solution. Hot and cold are antonyms.

The antonym of day is night.

Example 2. Pilot : Aeroplane :: Driver : ? (Road, Car, Engine, Licence)

Solution. A pilot operates an aeroplane.

A driver operates a car.

3.2.7 Foreign words & phrases

What it is: Expressions borrowed from other languages (mainly Latin and French) that are used in English.

How to identify it

- "What does 'bona fide' mean?" or choosing the right phrase for a given meaning.

How to solve it

- Learn the common list with meanings: bona fide (genuine), ad hoc (for this purpose only), status quo (the existing state), vice versa (the other way round), per capita (for each person), de facto (in fact), alma mater (one's former school), carte blanche (full freedom), déjà vu (a feeling of having seen something before), modus operandi (method of working), quid pro quo (something in return), persona non grata (an unwelcome person).
- Use them in your own sentences to remember them.

Example 1. What does "bona fide" mean? Use it in a sentence.

Solution. "Bona fide" means genuine or in good faith.

Example: "He is a bona fide customer with a valid receipt."

Example 2. What do "status quo" and "carte blanche" mean?

Solution. "Status quo" means the existing situation.

"Carte blanche" means full freedom to act as one wishes.

3.3 Sentence Structure & Flow

3.3.1 Fill in the blanks

What it is: Choosing the word or words that complete a sentence correctly in both meaning and grammar.

How to identify it

- Sentences with one or two blanks and four options.

How to solve it

- Read the whole sentence and spot the clue words: but, although, however (contrast); because, so, therefore (cause and result); and, also (same direction).
- Predict the missing word before reading the options.
- Check that the option fits the grammar and sounds natural with the words around it.
- With two blanks, test each pair together.

Example 1. Fill in the blank: "Although the plan seemed risky, it turned out to be highly _____. " (profitable / harmful / careless)

Solution. "Although" signals a contrast with "risky".

Answer: profitable.

Example 2. Fill in the blank: "The speaker was so _____ that the audience listened in complete silence." (boring / captivating / ordinary)

Solution. "So... that" links a cause to a result. Silent listening results from a gripping speaker.

Answer: captivating.

3.3.2 Cloze test

What it is: A passage with several blanks. Each blank has options, and the passage must make sense as a whole.

How to identify it

- A paragraph with numbered blanks and a set of options for each.

How to solve it

- Read the whole passage first to learn its subject and tone.
- Fill the easy blanks first (prepositions, articles, verb forms).
- For difficult blanks, use the sentences just before and after for meaning and for linking words.
- Re-read the completed passage to confirm it flows.

Example 1. "Reading is a habit that (1) _____ the mind. People who read daily (2) _____ a better vocabulary." (1) sharpens / closes / breaks. (2) has / have / having.

Solution. (1) The sentence praises reading, so "sharpens" fits.

(2) The subject "people" is plural, so "have".

Answer: sharpens, have.

Example 2. "She prepared well for the exam. (1) _____, she passed with distinction. (2) _____ her friend, who did not study, failed." (1) As a result / However. (2) Unlike / Similarly.

Solution. (1) Passing is the result of preparing, so "As a result".

(2) The friend's outcome is opposite, so "Unlike".

Answer: As a result, Unlike.

3.3.3 Para jumbles

What it is: Several sentences are given in the wrong order. You put them in the order that forms a logical paragraph.

How to identify it

- Sentences labelled P, Q, R, S (or 1 to 6) with the instruction "arrange in the correct order".

How to solve it

- Find the opening sentence: it introduces the subject and does not depend on an earlier sentence (no "this", "he", "however" referring back).
- Find mandatory pairs: a name must come before the pronoun that refers to it; "this" or "these" must follow the thing it points to; "however" follows the idea it contrasts with.
- Find the closing sentence: it concludes or looks ahead.
- Check the order for time or logic, then test the whole sequence by reading it.

Example 1. Arrange: P. He was awarded the first prize. Q. Ravi practised for months. R. As a result, he won the final. S. Ravi was selected for the district chess tournament.

Solution. S introduces Ravi. Q says he practised. R gives the result (winning the final). P follows (the prize).
Order: SQR P.

Example 2. Arrange: P. This led to a sharp rise in prices. Q. Heavy rain destroyed crops across the region. R. The government then released grain from its stock. S. The supply of vegetables fell.

Solution. Q is the cause. S is its effect. P ("This led to...") refers to the fall in supply. R is the reaction to the price rise.
Order: QSP R.

3.3.4 Odd sentence out

What it is: A paragraph has one sentence that does not belong to its theme. You find it.

How to identify it

- Five numbered sentences, with the instruction "find the sentence that does not fit".

How to solve it

- Find the central idea from the sentences that clearly connect.
- The odd sentence changes the topic, adds an unrelated detail or contradicts the flow.
- Test by reading the paragraph without the suspected sentence; if it reads smoothly, you have found it.

Example 1. Find the odd sentence: 1. Tea is grown in hilly regions. 2. Assam and Darjeeling are famous for it. 3. Cricket is India's most popular sport. 4. It needs plenty of rain. 5. India is among the largest tea producers.

Solution. Sentences 1, 2, 4 and 5 are about tea.
Sentence 3 is about cricket, so the odd sentence is 3.

Example 2. Find the odd sentence: 1. Regular exercise improves heart health. 2. It lowers stress and improves sleep. 3. Gold prices rose last week. 4. Even a 30-minute daily walk makes a difference.

Solution. Sentences 1, 2 and 4 are about the benefits of exercise.
Sentence 3 is about gold prices, so the odd sentence is 3.

3.3.5 Sentence insertion

What it is: A single sentence is given and you choose the position in a paragraph where it fits best.

How to identify it

- A paragraph with numbered gaps and the question "where should the sentence be placed?"

How to solve it

- Look at the given sentence for signals: pronouns, linking words (however, as a result, for example), time words.
- It must come after the sentence it refers to or contrasts with, and before the sentence that builds on it.
- Insert it in each position and read; keep the position where the flow is smooth.

Example 1. Insert the sentence "These subsidies made drip irrigation affordable." into the paragraph. [A] Groundwater levels in the region fell sharply. [B] Wells began to dry up. [C] The state promoted water-saving methods and gave subsidies. [D] The area under paddy decreased.

Solution. "These subsidies" must follow the sentence that first mentions subsidies, which is [C].
It goes after [C].

Example 2. Insert "However, the results were disappointing." into the paragraph. (1) The team tested the new software for six months. (2) It had been expected to double the speed of work. (3) Users found it too complicated to learn.

Solution. "However" contrasts with an earlier expectation, which appears in (2). The explanation of the disappointment comes in (3). It goes after (2) and before (3).

3.4 Comprehension

3.4.1 Reading comprehension

What it is: Reading a passage and answering questions on its main idea, details, inference, tone and vocabulary.

How to identify it

- A passage of 150 to 600 words followed by four to six questions.
- Question types: main idea or title, detail, inference, vocabulary in context, tone or attitude, "which statement is not true".

How to solve it

- Skim the questions first so that you know what to look for, then read the passage once at a steady pace, noting the point of each paragraph.
- Detail questions: find the line in the passage and match it.
- Inference questions: choose what must be true on the evidence, not what is merely possible.
- Vocabulary in context: put each option into the sentence and see which keeps the meaning.
- Eliminate extreme options (always, never, only, all) and options that are true in general but not stated in the passage.

Example 1. Passage: "Urban gardens are spreading in many cities. They provide fresh vegetables, reduce heat and bring neighbours together. However, they need regular care and access to water, which not every neighbourhood can provide." What is the main idea?

Solution. The passage lists the benefits of urban gardens and then the conditions they need.

The best choice is "Urban gardens are useful, but they depend on care and resources." An option like "Urban gardens are impossible" is too extreme.

Example 2. Using the same passage, which statement can be inferred? (a) Some neighbourhoods may find it hard to start urban gardens. (b) Every city resident grows vegetables.

Solution. The passage says not every neighbourhood can provide water, so some may struggle. This is supported. (b) is an extreme claim with no support.

Answer: (a).

3.4.2 Paragraph summary

What it is: Choosing the short statement that best sums up the main point of a paragraph.

How to identify it

- "Which of the following best summarises the paragraph?"

How to solve it

- Identify the main point and the key support for it.
- Pick the option that covers the whole paragraph, not just one detail.
- Reject options that add new facts, exaggerate, or focus on a small example.
- A good summary keeps the same tone and is much shorter than the paragraph.

Example 1. Choose the best summary: "Sleep is not a luxury. Lack of sleep reduces concentration, weakens immunity and raises the risk of long-term illness. Experts therefore advise seven to eight hours every night."

Solution. The main point is that sleep is essential, supported by health effects and a recommended duration.

Best summary: "Adequate sleep is essential for health, and experts recommend seven to eight hours." An option that mentions only concentration is too narrow.

Example 2. Choose the best summary: "Smartphones have made information available to millions of people. But constant notifications break attention and reduce productivity. Users who limit notifications report better focus."

Solution. The paragraph gives a benefit, a problem and a remedy.

Best summary: "Smartphones give easy access to information, but unless notifications are managed they distract users."

3.4.3 Critical reasoning (verbal)

What it is: Short passages that contain an argument. You identify the conclusion, an assumption, what strengthens or weakens it, or what can be inferred.

How to identify it

- "Which of the following, if true, weakens the conclusion?", "which assumption does the passage depend on?", "what can be inferred?"

How to solve it

- Mark the conclusion and the reasons given for it.
- Look for the gap between them: what must be true for the reasons to prove the conclusion?
- Weaken: show another cause, or that the link does not hold. Strengthen: support the link or remove other causes.
- Inference: choose what must be true using only the passage, not outside knowledge.

Example 1. "A school introduced a free breakfast programme. Attendance rose by 15% the following year. So the free breakfast raised attendance." Which weakens the conclusion most? (a) The school also began a strict attendance reward scheme the same year. (b) Students enjoy the breakfast.

Solution. (a) gives another possible cause of the rise, so the breakfast may not be the reason.

(b) is consistent with the claim and does not weaken it.

Answer: (a).

Example 2. For the same passage, which is the underlying assumption? (a) Attendance would not have risen by 15% without the breakfast programme. (b) All students attend school every day.

Solution. The conclusion needs the programme to be the cause. If attendance would have risen anyway, the argument fails. So (a) is the assumption.

(b) is not needed for the argument.

Part 4: Data Interpretation

Reading data from tables and charts and answering questions with quick calculation. Accuracy in reading the chart matters more than the arithmetic. Data in the examples is given in words.

4.1 Reading & Analysing Data

4.1.1 Tables

What it is: Data arranged in rows and columns, such as sales by year or marks by subject. Questions ask for totals, averages, percentages, ratios and comparisons.

How to identify it

- A grid of numbers followed by questions like "what percent increase", "in which year was the ratio highest", "what is the average".

How to solve it

- Read the title, the units (₹ lakh, thousands) and the row and column headings before anything else.
- Pick out only the cells the question needs. Do not calculate the whole table.
- Percentage change = $(\text{new} - \text{old}) \div \text{old} \times 100$. Compare ratios by cross-multiplying or by converting to decimals.
- Use approximations when the options are far apart.

Example 1. Sales of Product A (₹ lakh) were 40, 50 and 60 in 2022, 2023 and 2024. Find the percentage growth from 2022 to 2024 and the three-year average.

Solution. Growth = $(60 - 40) \div 40 \times 100 = 50\%$.

Average = $(40 + 50 + 60) \div 3 = ₹50$ lakh.

Example 2. Sales of Product B (₹ lakh) were 30, 36 and 45 in the same years. In which year was the ratio of A's sales to B's sales highest?

Solution. 2022: $40 \div 30 = 1.333$. 2023: $50 \div 36 = 1.389$. 2024: $60 \div 45 = 1.333$.

The ratio was highest in 2023.

4.1.2 Bar graphs

What it is: Bars whose heights show values for different categories or years. There may be double bars (two series side by side) or stacked bars (parts piled on top of each other).

How to identify it

- Rectangular bars on labelled axes, with questions comparing bars or calculating from their values.

How to solve it

- Check what each gridline represents (the scale) and the units.
- Read each bar carefully and write the values in a small table before calculating.
- For stacked bars, the value of a segment is the difference between its top and bottom.
- Then calculate the quantity asked: difference, percent change, ratio or average.

Example 1. A bar graph shows production (thousand tonnes): Jan 40, Feb 55, Mar 45, Apr 60. In which month was the percentage increase over the previous month the highest?

Solution. Feb: $(55 - 40) \div 40 = 37.5\%$. Mar: a fall. Apr: $(60 - 45) \div 45 = 33.3\%$.

The highest increase was in February (37.5%).

Example 2. A double bar graph shows the number of passed students (in hundreds): School X: boys 30, girls 20. School Y: boys 25, girls 35. School Z: boys 40, girls 30. Find the ratio of total girls to total boys.

Solution. Girls = $20 + 35 + 30 = 85$. Boys = $30 + 25 + 40 = 95$.

Ratio = $85 : 95 = 17 : 19$.

4.1.3 Line graphs

What it is: Points joined by lines to show how a quantity changes over time, so trends (rise, fall, peak) are easy to see.

How to identify it

- A line with points over years or months, with questions on rise and fall, growth rate or comparison of two lines.

How to solve it

- Read the value at each point using the axis scale, and write them down.
- The steepest segment shows the biggest change per period, but the biggest percentage change also depends on the starting value, so calculate it.
- Percentage change between two points = $(\text{difference} \div \text{starting value}) \times 100$.
- Where two lines cross, the two quantities are equal at that point.

Example 1. A line graph shows profit (₹ lakh) from 2020 to 2024: 20, 24, 30, 27, 36. In which year was the percentage rise the highest?

Solution. 2021: $4 \div 20 = 20\%$. 2022: $6 \div 24 = 25\%$. 2023: a fall. 2024: $9 \div 27 = 33.3\%$.
The highest rise was in 2024 (33.3%).

Example 2. Company P's sales were 50, 60 and 70, and Company Q's were 40, 55 and 75, in three successive years. In which year did Q first exceed P, and what were the totals?

Solution. Year 1: $40 < 50$. Year 2: $55 < 60$. Year 3: $75 > 70$.
Q first exceeded P in year 3. Totals: P = 180 and Q = 170.

4.1.4 Pie charts

What it is: A circle divided into sectors. Each sector's angle or percentage shows its share of the whole.

How to identify it

- A circular chart labelled with percentages or degrees, with a total given separately.

How to solve it

- The whole circle is 360° , which is 100%. So $1\% = 3.6^\circ$.
- Value of a sector = $(\text{angle} \div 360) \times \text{total}$, or $(\text{percent} \div 100) \times \text{total}$.
- Compare sectors using their angles or percentages. If two pies have different totals, convert to actual values before comparing.
- Check that the percentages add up to 100.

Example 1. A company spends ₹7.2 lakh in a year. The sector for salaries has an angle of 126° . What percent and what amount is spent on salaries?

Solution. Share = $126 \div 360 = 35\%$.
Amount = $0.35 \times 7.2 = ₹2.52$ lakh.

Example 2. A family's budget pie shows Food 30%, Rent 25%, Education 20%, Savings 15% and Others 10%. The income is ₹60,000. Find the angle for Education and the amount spent on Rent.

Solution. Education angle = $20\% \times 360^\circ = 72^\circ$.
Rent = $25\% \times 60,000 = ₹15,000$.

4.1.5 Mixed or combined graphs

What it is: Two or more charts used together (bar with line, table with pie chart). Each question needs values from more than one chart.

How to identify it

- Two charts side by side, or one chart that supplies totals and another that supplies percentages.

How to solve it

- Write the data from each chart into one small table.
- Check units: one chart may be in thousands and another in lakhs.
- Convert percentages into actual values (using the total from the other chart) before you compare or add.
- Work in steps, one derived quantity at a time.

Example 1. A bar chart shows total students: School A 800, School B 600. A pie chart shows the percentage of girls: School A 55%, School B 45%. Find the total number of girls in the two schools.

Solution. School A girls = $0.55 \times 800 = 440$. School B girls = $0.45 \times 600 = 270$.
Total = $440 + 270 = 710$.

Example 2. A table gives total expenses: A ₹50,000 and B ₹80,000. A pie chart shows travel as 20% of A's expenses and 15% of B's. Who spent more on travel, and by how much?

Solution. A: 20% of 50,000 = ₹10,000. B: 15% of 80,000 = ₹12,000.
B spent ₹2,000 more.

4.1.6 Radar or web charts

What it is: A chart with several axes radiating from the centre, one for each category. The values on the axes are joined to form a closed shape, which shows a profile.

How to identify it

- A spider-web shaped chart, often comparing two or more people or products across the same set of categories.

How to solve it

- Each spoke has its own scale that starts at the centre. Read the value where the line crosses each spoke.
- Points farther from the centre mean larger values.
- Write the readings in a table (category against entity) and then do the calculation.
- To compare two entities, compare spoke by spoke.

Example 1. A radar chart shows Student X's scores out of 10: Maths 8, Science 6, English 7, History 5, Art 9. Find the average score and the strongest subject.

Solution. Total = $8 + 6 + 7 + 5 + 9 = 35$.
Average = $35 \div 5 = 7$. The strongest subject is Art (9).

Example 2. Student Y's scores are Maths 6, Science 8, English 7, History 6, Art 8. In how many subjects did X score higher than Y?

Solution. Maths: $8 > 6$ (X higher). Science: $6 < 8$. English: $7 = 7$. History: $5 < 6$. Art: $9 > 8$ (X higher).
X scored higher in 2 subjects.

4.1.7 Caselets

What it is: Data given as a paragraph of text rather than a chart. You turn it into a table or diagram first.

How to identify it

- A paragraph full of numbers, percentages and ratios, followed by several questions.

How to solve it

- Read the paragraph twice and underline each number and relation.
- Build a table (or a Venn diagram or tree) with a box for every quantity, putting unknowns as letters.
- Fill in what is known, then use ratios and percentages to find the rest, one box at a time.
- Answer the questions from the finished table.

Example 1. A college has 500 students, 60% of whom are boys. 40% of the boys and 50% of the girls play a sport. How many students play a sport?

Solution. Boys = $0.6 \times 500 = 300$. Girls = 200.
Boys who play = $0.4 \times 300 = 120$. Girls who play = $0.5 \times 200 = 100$.
Total = 220 students.

Example 2. A shop sold 120 items: pens, notebooks and files in the ratio 3 : 4 : 5. Pens cost ₹10, notebooks ₹20 and files ₹15. Find the total revenue.

Solution. Total parts = 12 and one part = $120 \div 12 = 10$ items.
Pens = 30, notebooks = 40, files = 50.
Revenue = $30 \times 10 + 40 \times 20 + 50 \times 15 = 300 + 800 + 750 = ₹1,850$.

4.1.8 Venn-based data interpretation

What it is: Data given using overlapping groups (circles), or in sentences that describe overlaps, such as how many like tea, coffee, or both.

How to identify it

- A Venn diagram, or statements giving sizes of groups and their overlaps.

How to solve it

- Fill the innermost overlap first, then work outwards.
- Only A = $n(A) - \text{overlaps with A}$. Total in at least one group = $n(A) + n(B) - n(A \cap B)$ (extend for three sets).
- Neither = total – at least one.
- When percentages are given, either convert to numbers using the total or work in percentages throughout.

Example 1. In a survey of 200 people, 120 like tea, 90 like coffee and 40 like both. How many like only tea, only coffee, and neither?

Solution. Only tea = $120 - 40 = 80$. Only coffee = $90 - 40 = 50$.

At least one = $120 + 90 - 40 = 170$.

Neither = $200 - 170 = 30$.

Example 2. Of 100 employees, 60% know Java, 50% know Python and 20% know neither. What percentage know both?

Solution. At least one language = $100\% - 20\% = 80\%$.

$80 = 60 + 50 - \text{both}$, so both = $110 - 80 = 30\%$.

4.1.9 Missing data

What it is: A table or chart has blank cells. You find the missing values from totals, averages, ratios or percentages given elsewhere.

How to identify it

- A table with "?" or blanks, together with clues such as "the total is..." or "the average is...".

How to solve it

- Use row and column totals to find a blank when only one value is missing.
- Use given averages (total = average $\times$ count), ratios and percentages to link unknown cells.
- Solve in an order that gives one unknown at a time.
- Put the values back in and check them against every total.

Example 1. Regional sales (₹ lakh): North 25, South ?, East 18, West 22. The total is 90. Find South's sales and its share of the total.

Solution. South = $90 - (25 + 18 + 22) = 90 - 65 = 25$.

Share = $25 \div 90 = 27.8\%$.

Example 2. The average of four quarterly sales is ₹55 lakh. Q1 is 40, Q2 is 50 and Q4 is 70. Find Q3.

Solution. Total = $4 \times 55 = 220$.

Known quarters = $40 + 50 + 70 = 160$.

Q3 = $220 - 160 = ₹60$ lakh.

4.1.10 Data sufficiency (data interpretation)

What it is: A question about data is followed by statements. You decide whether the statements give enough information to find a single answer.

How to identify it

- "Is the data in statement I alone sufficient to answer the question?" with options such as "both statements are required".

How to solve it

- For each statement ask: can I get the exact quantity asked for?
- Percentages without a base value cannot give actual amounts. Ratios without any actual figure give only ratios.
- Combine the statements only if neither works alone.
- Quick test: count the unknowns and the independent facts; you need at least as many facts as unknowns.

Example 1. What is the number of girls in a class? I. Girls are 40% of the class. II. The class has 50 students.

Solution. I alone gives a percentage but no total.

II alone gives a total but not how many are girls.

Together: 40% of 50 = 20. Both statements are needed.

Example 2. What is the percentage growth in revenue from 2023 to 2024? I. Revenue in 2023 was ₹80 lakh. II. Revenue in 2024 was 25% more than in 2023.

Solution. II directly gives the growth as 25%.

I gives only the 2023 amount and no information about 2024.

Statement II alone is sufficient.

4.1.11 Core calculation skills

What it is: The calculations used again and again in data interpretation: percentage change, share, ratio, average, percentage points and compound annual growth rate (CAGR).

How to identify it

- Words such as growth, share, contribution, percentage points, CAGR, average annual increase.

How to solve it

- Percentage change = $(\text{new} - \text{old}) \div \text{old} \times 100$. Share = $\text{part} \div \text{total} \times 100$.
- Percentage points compare two percentages by subtraction. Going from 20% to 25% is +5 percentage points, which is a 25% increase.
- Average = $\text{sum} \div \text{count}$. Use a weighted average when the groups have different sizes.
- $\text{CAGR} = (\text{final} \div \text{initial})^{1/n} - 1$, where n is the number of years between the two values (not the number of data points).
- Learn fraction equivalents ($1/8 = 12.5\%$, $1/6 = 16.7\%$) and compare fractions by cross-multiplication.

Example 1. Revenue grew from ₹100 crore to ₹121 crore in 2 years. Find the CAGR.

Solution. $\text{CAGR} = (121 \div 100)^{1/2} - 1 = 1.1 - 1$.

CAGR = 10% per year.

Example 2. A company's market share rose from 20% to 25%. State the change in percentage points and in percent.

Solution. Percentage points: $25 - 20 = 5$.

Percent: $(25 - 20) \div 20 \times 100 = 25\%$ increase.

Part 5: Technical Aptitude

Common in IT and campus placement tests: core computer science concepts and output prediction. These questions reward clear fundamentals and careful tracing of code.

5.1 Core Computer Science

5.1.1 Computer fundamentals

What it is: The basic ideas about how a computer works: hardware, software, memory, units of storage and basic networking.

How to identify it

- "Which of the following is volatile memory?", "1 KB equals...", "the brain of a computer", "full form of..."

How to solve it

- CPU = ALU (calculations and logic) + control unit + registers.
- Memory hierarchy from fastest to slowest: registers, cache, RAM, secondary storage (hard disk, SSD). Faster memory is smaller and costs more.
- RAM is volatile (loses data when power goes off). ROM and storage devices are non-volatile.
- 1 byte = 8 bits. 1 KB = 1024 bytes. 1 MB = 1024 KB. 1 GB = 1024 MB.
- Software types: system software (operating system, drivers), application software, utility software.

Example 1. Which of these is volatile memory: ROM, RAM, hard disk, flash drive?

Solution. RAM loses its contents when the power is switched off. ROM, the hard disk and a flash drive keep their data.

Answer: RAM.

Example 2. How many bits are there in 2 KB?

Solution. 2 KB = 2×1024 bytes = 2048 bytes.
Bits = $2048 \times 8 = 16,384$.

5.1.2 Programming & output prediction

What it is: Reading a short program and saying what it prints, by following it line by line.

How to identify it

- A code snippet in C, C++, Java or Python followed by "What is the output?"

How to solve it

- Keep a trace table: write each variable's value after every line or loop pass.
- Know the operators: integer division discards the fraction in C and Java, % gives the remainder, ++ and -- change the value by 1, && and || stop early (short-circuit).
- In Python, / gives a decimal result and // gives the integer quotient. range(a, b) stops before b.
- Check off-by-one errors in loops and the exact loop condition.

Example 1. Java: `int a = 10, b = 3; System.out.println(a / b + a % b);` What is printed?

Solution. $a / b = 3$ (integer division). $a \% b = 1$.
 $3 + 1 = 4$. The output is 4.

Example 2. Python: `s = 0; for i in range(1, 5): s += i * i; print(s)`. What is printed?

Solution. range(1, 5) gives $i = 1, 2, 3, 4$.
 $s = 1 + 4 + 9 + 16 = 30$. The output is 30.

5.1.3 Data structures & algorithms

What it is: Data structures organise data (array, linked list, stack, queue, tree, graph, hash table). Algorithms are step-by-step methods such as searching and sorting. Their cost is described by time complexity.

How to identify it

- "Which data structure follows LIFO", "time complexity of binary search", "inorder traversal", "number of comparisons".

How to solve it

- Stack: last in, first out (push and pop). Queue: first in, first out.
- Linear search: $O(n)$. Binary search (sorted data): $O(\log n)$. Bubble, selection and insertion sort: $O(n^2)$. Merge sort and heap sort: $O(n \log n)$. Quick sort: $O(n \log n)$ on average, $O(n^2)$ worst case.
- Binary tree traversals: inorder (left, root, right), preorder (root, left, right), postorder (left, right, root). The inorder traversal of a binary search tree is sorted.
- BFS uses a queue. DFS uses a stack or recursion.
- Binary search on n items needs at most $\lfloor \log_2 n \rfloor + 1$ comparisons.

Example 1. Insert 50, 30, 70, 20, 40 into an empty binary search tree. What is the inorder traversal?

Solution. 50 is the root. 30 goes left of 50, 70 goes right. 20 goes left of 30 and 40 goes right of 30. Inorder (left, root, right): 20, 30, 40, 50, 70, which is sorted.

Example 2. What is the maximum number of comparisons in a binary search of 1000 sorted items?

Solution. $\lfloor \log_2 1000 \rfloor = 9$, because $2^9 = 512$ and $2^{10} = 1024$.
Comparisons = $9 + 1 = 10$.

5.1.4 OOP concepts

What it is: Object-oriented programming organises code into classes and objects, using encapsulation, inheritance, polymorphism and abstraction.

How to identify it

- "Which concept hides data", "what is the output when an overridden method is called", "overloading versus overriding".

How to solve it

- Encapsulation: data and methods bundled together, with data hidden using private access.
- Inheritance: a child class reuses and extends a parent class.
- Polymorphism: one name, many forms. Overloading (same method name, different parameters, same class) is decided at compile time. Overriding (same signature in a child class) is decided at run time by the actual object type.
- Abstraction: show what an object does and hide how (abstract classes and interfaces).
- A constructor builds the object. "this" refers to the current object, "super" to the parent.

Example 1. Java: `class A { void show() { System.out.println("A"); } }` `class B extends A { void show() { System.out.println("B"); } }` `A obj = new B(); obj.show();` What is printed?

Solution. B overrides show(). The method that runs depends on the actual object (a B), not on the reference type. The output is B (run-time polymorphism).

Example 2. A class has two methods, `add(int a, int b)` and `add(double a, double b)`. Which OOP concept is this?

Solution. The same method name is used with different parameter types in one class. This is method overloading (compile-time polymorphism).

5.1.5 DBMS & SQL

What it is: A database management system stores data in tables. SQL is the language used to create, read, update and delete that data.

How to identify it

- "What does this query return?", "which normal form", "primary key versus foreign key", "type of join".

How to solve it

- A primary key uniquely identifies each row. A foreign key links to the primary key of another table.
- Query layout: SELECT ... FROM ... WHERE ... GROUP BY ... HAVING ... ORDER BY. They run in the order FROM, WHERE, GROUP BY, HAVING, SELECT, ORDER BY. WHERE filters rows before grouping; HAVING filters groups after.
- Joins: INNER (only matching rows), LEFT (all left rows plus matches), RIGHT (all right rows plus matches), FULL (all rows from both).
- Aggregate functions: COUNT, SUM, AVG, MAX, MIN.
- Normalisation: 1NF (atomic values), 2NF (no partial dependency on part of a key), 3NF (no transitive dependency).

Example 1. Table Emp(id, name, dept, salary). What does this query return? SELECT dept, COUNT(*) FROM Emp GROUP BY dept HAVING COUNT(*) > 2;

Solution. It groups employees by department, counts each group, and keeps only groups with more than 2 employees.

It returns each department that has more than 2 employees, with its employee count.

Example 2. Table A has ids 1, 2, 3 and table B has ids 2, 3, 4. How many rows do A INNER JOIN B ON A.id = B.id and A LEFT JOIN B ON A.id = B.id return?

Solution. Inner join keeps only matching ids 2 and 3, so 2 rows.

Left join keeps all of A (ids 1, 2, 3), so 3 rows. The row for id 1 has NULL for B's columns.

5.1.6 Operating systems

What it is: The operating system manages processes, memory, files and devices. Typical questions are on CPU scheduling, deadlocks and paging.

How to identify it

- "Average waiting time under FCFS", "which scheduling is pre-emptive", "conditions for deadlock", "number of page faults".

How to solve it

- Scheduling: FCFS runs processes in order of arrival; SJF picks the shortest burst time; Round Robin gives each process a fixed time quantum. Waiting time = start time – arrival time (when there is no pre-emption).
- Deadlock needs all four conditions: mutual exclusion, hold and wait, no pre-emption, circular wait.
- Paging: the logical address is translated to a physical address using a page table. A page fault happens when the needed page is not in memory. Replacement policies: FIFO, LRU, Optimal.
- Process states: new, ready, running, waiting, terminated.

Example 1. Three processes P1, P2 and P3 arrive at time 0 with burst times 6, 3 and 1. Find the average waiting time under FCFS.

Solution. P1 starts at 0, so waits 0. P2 starts at 6, so waits 6. P3 starts at 9, so waits 9.

Average = $(0 + 6 + 9) \div 3 = 5$.

Example 2. Using FIFO page replacement with 3 frames, how many page faults occur for the reference string 1, 2, 3, 1, 4, 2, 1?

Solution. 1, 2, 3 are faults (frames 1, 2, 3). 1 is a hit. 4 is a fault and replaces the oldest page, 1 (frames 4, 2, 3). 2 is a hit. 1 is a fault and replaces the oldest page, 2 (frames 4, 1, 3).

Faults = $3 + 1 + 1 = 5$.

5.1.7 Computer networks

What it is: How computers communicate with each other, covering the OSI and TCP/IP layers, protocols and IP addressing.

How to identify it

- "At which layer does a router work", "default port of HTTPS", "number of hosts in a /26 subnet", "TCP versus UDP".

How to solve it

- OSI layers from top to bottom: Application, Presentation, Session, Transport, Network, Data Link, Physical.
- Network layer: IP addresses and routers. Data Link layer: MAC addresses and switches. Transport layer: TCP and UDP.
- TCP is reliable and connection-oriented. UDP is faster but does not guarantee delivery.
- Well-known ports: HTTP 80, HTTPS 443, FTP 21, SSH 22, SMTP 25, DNS 53.
- IPv4 addresses have 32 bits. A /n prefix leaves $32 - n$ host bits, giving $2^{32-n} - 2$ usable host addresses.

Example 1. How many usable host addresses are there in a /26 network?

Solution. Host bits = $32 - 26 = 6$.

Usable hosts = $2^6 - 2 = 62$.

Example 2. At which OSI layer does a router work, and which protocol turns a domain name into an IP address?

Solution. A router forwards packets using IP addresses, so it works at the Network layer (layer 3).

The Domain Name System (DNS) translates names into IP addresses.

5.1.8 Logical & pseudocode questions

What it is: Reading simple pseudocode (loops, conditions, recursion) and working out its result without a computer.

How to identify it

- "Consider the following pseudocode... what is the value of x?", "what does the function return?"

How to solve it

- Make a trace table with one column for each variable and one row for every loop iteration.
- Apply the loop condition before every pass and the update after every pass.
- Be careful with integer division and with the order of statements inside the loop.
- Recursion: expand the calls from the base case upward, writing each returned value.

Example 1. Pseudocode: $x = 1$; $y = 10$; while $y > 1$: $x = x \times 2$; $y = y \div 2$ (integer division). What is x at the end?

Solution. Pass 1: $y = 10 > 1$, so $x = 2$, $y = 5$. Pass 2: $y = 5 > 1$, so $x = 4$, $y = 2$. Pass 3: $y = 2 > 1$, so $x = 8$, $y = 1$. $y = 1$ is not greater than 1, so the loop stops. $x = 8$.

Example 2. Define $f(n) = n$ if $n \leq 1$, otherwise $f(n - 1) + f(n - 2)$. Find $f(6)$.

Solution. $f(0) = 0$, $f(1) = 1$, $f(2) = 1$, $f(3) = 2$, $f(4) = 3$, $f(5) = 5$.

$f(6) = f(5) + f(4) = 5 + 3 = 8$.